INTENTIONAL SEMANTIC GEOMETRY: STATEMENTS AS GRAMMAR-MEDIATED REFERENCES INTO DETERMINABLE SPECTRA

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Abstract. Non-trivial scientific meaning is not, in general, a proposition in a fixed logic-space. A statement is a *pointer* into a semantic geometry — a *spectrum* Σ — and its interpretation is *grammar-mediated*: a grammar G decides how the statement names regions or positions in Σ. Both the pointing (*where* in Σ does the statement land?) and the substrate (*what is* Σ, structurally?) are objects of research. We propose Intentional Semantic Geometry (ISG) as the discipline that makes this structure explicit: statements, grammars, spectra, references, and two irreducible modes of indeterminacy (*pointing* indeterminacy under a fixed substrate, *substrate* indeterminacy across candidate grammars). Under ISG, the operators DIA (which names the interpretive gap left by a partition) and REI (which reintegrates that gap) recover a natural position as the derived discipline for handling the double indeterminacy. We equip them with a typed calculus that separates the type of an operation from the direction in which it is applied and from its normative valence, and that identifies which reintegration routes transform a state and which only certify one. The same calculus types the two deployment effects of a separating criterion — excisive *cutting* and non-excisive *classification*, which refines the description of an intact spectrum — and states the typed interface through which classified, still-unnamed regions become inputs to symbolic reification in the companion MCK framework. The framework unifies substrates already studied independently in the programme (topological property classes over infinite traces; epistemic region cubes; capability lattices for threshold monitors; two-agent channel-based semantic bodies), and provides the language for a reflexive application: a paper's positioning statements about its own contribution are ISG operations on the contribution's admissible-readings spectrum, subject to the same double indeterminacy as any other statement about a geometrically-supported object.

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Contents

1. Introduction 3
1.1. The problem 3
1.2. Contribution 3
1.3. Roadmap 4
2. The Framework: Statements, Grammars, Spectra 5
2.1. Statements 5
2.2. Spectra 5
2.3. Grammars and references 5
2.4. Two indeterminacy modes 6
3. Solid Integral, Atomic Geometry, and the SP–SR–IE Canon 6
3.1. Two modes of specifying a spectrum 6
3.2. The SP–SR–IE canon 8
4. The Generative Cycle: from Constitutive DIA through Symbolic Genesis to REI Discipline 9
4.1. Constitutive DIA 10
4.2. Grammar-selection DIA (meta-distinction) 11
4.3. Symbolic genesis 11
4.4. Pointing DIA 15
4.5. Substrate DIA 16
4.6. REI: four routes to reintegration 16
4.7. A typed calculus for DIA and REI 18
4.8. A theorem-target: DIA-freedom and topological clopenness 28
4.9. The aftermath of the constitutive cut: the (M_1, C, M_2) separation 29
5. Programme Substrates as ISG Instances 31
5.1. ADKL hyperspheres (Solid Integral) 31
5.2. The MCK region cube (Atomic Geometry) 32
5.3. The S/C/L/K four-family (Atomic Geometry) 32
5.4. Touch and Crossing (dual Atomic Geometry) 32
5.5. Semantic body: two-agent, three-channel substrate 32
5.6. Epistemic body: shared witness register under substrate DIA 33
5.7. Two-register dynamics: ISG in motion 34
6. Reflexive ISG: Statements about a Scientific Contribution 35
6.1. Setup 35
6.2. Scope: truth-apt positioning 35
6.3. Positioning statements as ISG operations 35
6.4. Independence from evaluation 36
6.5. Payoff for programme papers 36
7. Related Work 37
8. Discussion, Open Questions, Future Work 39
8.1. Immediate applications 39
8.2. Open structural questions 39
8.3. Grounding for downstream programme work 40

References 40

Convention. Draft status. This manuscript is the P0 foundational paper of a coordinated cluster whose companion papers are in parallel preparation (applications, epistemic-body substrate, integrate-and-fire two-register dynamics). Cross-references to those companions are made explicit and their status (deposited / manuscript-in-preparation) is signalled at the citation.

1. Introduction

1.1. **The problem.** Consider a decomposition of an infinite-trace property φ into finitely verifiable pieces $\varphi_1, \ldots, \varphi_k$; a bounded model-checking run that reports "no counterexample within depth k"; a k-liveness weakening of $\mathsf{GF}p$; a CEGAR pass that discards concrete detail; a Tseitin encoding of a Boolean circuit; a finite-prefix runtime monitor. In every case the operational report tacitly claims that the enumerated pieces $\{A_1, \ldots, A_n\}$ exhaust a semantically-relevant domain D. Yet in every case there is a residual: an interval $E \subseteq D$ that no A_i names, that the report does not acknowledge, and that carries the very semantic content the decomposition was meant to expose. The word *diástēma* (*diástēma*, "interval, gap") captures this residual.

The residual is not a mistake — it is the natural cost of forcing a finite grammar onto a substrate the finite grammar cannot fully name. Refusing to acknowledge it, however, is a mistake: it converts an accountably-partial statement into an inflated total one.

The residual behaviour is also not confined to verification. It arises whenever a statement refers into a substrate that is itself under determination: a semantic embedding whose axes we choose; a monitor whose verdict alphabet we fix; a symbolic representation whose grammar we agree upon; a scientific paper whose contribution the author names under a chosen frame. Naming an interpretation *selects* regions on a substrate whose full structure has not been given.

1.2. **Contribution.** We propose *Intentional Semantic Geometry* (ISG) as the framework that makes the following structure explicit and computationally tractable.

(1) A statement s is an uninterpreted syntactic pointer.
(2) A *spectrum* Σ is the semantic geometry the statement refers into: a structured space carrying admissibility and exclusion relations. Substrates from across the programme — topological cells over Σ^ω; the MCK epistemic cube; the touch/crossing capability lattice; two-agent channel spaces — are all spectra.
(3) A *grammar* G mediates the interpretation, assigning to s a *reference* $\rho_G(s) \subseteq \Sigma$: the region of Σ the statement names.
(4) Two irreducible modes of indeterminacy sit inside this structure: *pointing indeterminacy* (given G and Σ, the excluded interval left by $\rho_G(s)$ inside Σ) and *substrate indeterminacy* (across candidate grammars $G_1, G_2, \ldots$, the excluded interval between the substrates $\Sigma_1, \Sigma_2, \ldots$ they determine).

(5) The operators DIA (naming the interpretive gap) and REI (reintegrating it) are the derived discipline for both modes.
(6) Those operators obey a *typed calculus* (Section 4.7). The type of an operation, the direction in which it is applied — at the carrier, or at the constraints presenting admissibility — and its normative valence are three axes. They are independent away from the constitutive stage, where the constraint-directed reading is not yet available, and — for the claim that the operator does not fix the direction — under a stated availability hypothesis on the presentation; several adapters restrict the pairing further. The separating criterion is a datum distinct from the operator that installs it; one reintegration route certifies without transforming; and no route inverts the cut that preceded it.
(7) A separating criterion does not determine what its application does to the spectrum: the same criterion can be deployed *excisively* (a cut: material moves to the excluded component, with boundary and provenance) or *non-excisively* (a classification: the admissible family is refined while every cell remains admitted and the excluded component is untouched). The deployment mode is an input of the act, not a property of the criterion (Section 4.7.6); classification feeds, across a typed interface with strict level separation, the symbolic reification machinery of the companion MCK embedding framework [Bra+26].

The framework has two poles, both inherited from earlier programme work [Bra26c]: *Solid Integral* (SI), the extensional mode in which Σ is given as measurable regions, and *Atomic Geometry* (AG), the intensional mode in which Σ is generated from atoms and constructive combinatorial rules. The classical clopen $\Leftrightarrow$ monitorable characterisation [Bra26e] becomes, under ISG, an Atomic-Geometry structure theorem; the ADKL hypersphere realisation [Bra26a] becomes a Solid-Integral realisation of the same substrate.

The reflexive application makes explicit a step scientific writing already performs implicitly: when a paper says "this contribution is K" or "this is not K", it partitions the space of admissible readings of its own contribution — a spectrum that is itself grammar-mediated and not fully determined. The positioning statement is a double-indeterminacy DIA on that spectrum. Its correctness, usefulness, or venue-appropriateness are subsequent REI-evaluations, not conditions for the classification. Downstream programme papers can cite this reflexive result to separate their invariant scientific core (definitions, assumptions, proofs, falsifiable scope conditions — H2/H4/H6-invariants of the programme) from their optional interpretive positioning (disciplinary label, excluded neighbouring frame, venue voice), without conflating the two.

1.3. **Roadmap.** Section 2 introduces the framework: statement, spectrum, grammar, reference, and the two indeterminacy modes. Section 3 recovers the two ISG modes (Solid Integral, Atomic Geometry) and the sub-canon of *admitted region* SP, *forbidden region* SR, and *excluded interval* IE. Section 4 recasts DIA and REI as the derived discipline for handling the two indeterminacy modes and states a clopen-based theorem-target for DIA-freedom. Section 5 exhibits programme substrates as ISG instances. Section 6 works out the reflexive application to scientific-contribution positioning. Section 7 places the framework against topos-theoretic, sheaf-theoretic, and category-theoretic neighbours. Section 8 lists open questions and future companion work.

2. The Framework: Statements, Grammars, Spectra

2.1. **Statements.**

Definition 1 (Statement)**.** A *statement* s is an element of a fixed alphabet $\mathcal{S}$, treated as an uninterpreted syntactic token. We make no prior commitment to the internal structure of s (formula, sentence, word, token stream). Interpretation is deferred to a grammar (Definition 4).

The deliberate absence of Boolean truth-value at Definition 1 is not an oversight; it is the framework's starting move. In ISG, s acquires content only through a grammar that maps it into a spectrum. Without a grammar and a spectrum, a statement is a pointer with no target.

2.2. **Spectra.**

Definition 2 (Spectrum)**.** A *spectrum* is a triple $\Sigma = (X, \mathsf{Adm}, \mathsf{Exc})$ where:

- X is a set (the *support*), equipped with at least the structure of a topological space; special cases will add metric, measure, order, category-theoretic, or graph structure as needed by the instance.
- $\mathsf{Adm} \subseteq \mathcal{P}(X)$ is a family of *admissible regions* — subsets of X into which a grammar can legitimately land a reference.
- $\mathsf{Exc} \subseteq \mathcal{P}(X)$ is a family of *explicitly excluded regions* — subsets a grammar may name as forbidden.

No algebraic-closure conditions are imposed on Adm and Exc at the framework level: they are families, not required to form a lattice, algebra, or σ-algebra. Specific instances may add structure — Solid-Integral spectra (Definition 9) typically require closure under finite Boolean combinations of primitive regions; Atomic-Geometry spectra (Definition 10) generate Adm constructively from atoms plus generation rules. Downstream operators that use set operations on references (Definition 6) do not require Adm or Exc to be closed under those operations, since the operations are always available on $\mathcal{P}(X)$. The pair $(\mathsf{Adm}, \mathsf{Exc})$ is not required to cover $\mathcal{P}(X)$: the residual $X \setminus (\bigcup \mathsf{Adm} \cup \bigcup \mathsf{Exc})$ is the substrate's own *under-determined interval*, and its identification is part of what a grammar is supposed to do.

Remark 3. The definition is deliberately minimal. Instances (Section 5) will impose additional structure — Borel σ-algebra, metric, lattice ordering, category with pullbacks — as the substrate demands. What matters at the framework level is that a spectrum comes equipped with a distinguished admissibility structure separate from its ambient topology; and that this structure may be incomplete.

2.3. **Grammars and references.**

Definition 4 (Grammar; reference)**.** A *grammar* for a statement space $\mathcal{S}$ into a spectrum $\Sigma = (X, \mathsf{Adm}, \mathsf{Exc})$ is a (partial) function $G : \mathcal{S} \rightharpoonup \mathsf{Adm} \cup \mathsf{Exc}$. Given s in the domain of G, the value $\rho_G(s) := G(s) \subseteq X$ is the *reference* of s under G: the region of the spectrum G says s names. When $\rho_G(s) \in \mathsf{Adm}$ we say G *admits* s; when $\rho_G(s) \in \mathsf{Exc}$, that G *excludes* s.

Two aspects of Definition 4 carry the framework's weight. First, G is a *function*: it commits to a specific region per statement, not a collection of candidates. Ambiguity, when it arises, is captured downstream by comparing different grammars $G_1, G_2, \ldots$ over the same s. Second, G is *partial*: statements outside its domain are not automatically false, they are simply unaddressed under the current grammar — a distinction the substrate indeterminacy mode (Section 2.4) will exploit.

Example 5 (Toy). Let $\mathcal{S} = \{s_1, s_2\}$ with s_1 = "the property holds eventually" and s_2 = "the property holds forever after some point". Let Σ have $X = \{\mathsf{S}, \mathsf{C}, \mathsf{L}, \mathsf{K}\} \cup \{\text{intersections}\}$ (a 16-cell partition; see Section 5). A grammar G_{LTL} that reads s_1 under a standard LTL semantics gives $\rho_{G_{\mathrm{LTL}}}(s_1) = \mathsf{C} \cap \mathsf{L}$ (cosafety and liveness). An alternative grammar G_{fair} that reads s_1 under a fairness scheme returns $\mathsf{L} \cap \mathsf{K}$ (recurrence). The two grammars give different references for the same statement; neither is intrinsically wrong. The disagreement is a substrate-indeterminacy datum, not an error.

2.4. **Two indeterminacy modes.** The framework's characteristic move is to split indeterminacy into two irreducible modes.

Definition 6 (Pointing indeterminacy)**.** Fix a spectrum Σ and a grammar G. The *pointing indeterminacy* of s under (G, Σ) is the residual

$$E_{\mathrm{point}}(s; G, \Sigma) \;:=\; X \setminus \big(\rho_G(s) \cup \rho_G(\neg s)\big)$$

— the set of positions in Σ that G neither admits under s nor explicitly excludes via $\neg s$. (If G does not treat negation, use $\rho_G(s)$ and $X \setminus \rho_G(s)$ as the two sides.)

Definition 7 (Substrate indeterminacy)**.** Fix a statement s and a family $\mathcal{F} = \{(G_i, \Sigma_i)\}_{i \in I}$ of candidate (grammar, spectrum) pairs each treating s. The *substrate indeterminacy* of s across $\mathcal{F}$ is the family of ordered pairs $(i, j) \in I \times I$ for which no reference-preserving translation between (G_i, Σ_i) and (G_j, Σ_j) at s exists in the sense to be made precise in Definition 28. The formal object built from this family is $\mathrm{DIA}_{\mathrm{sub}}(s; \mathcal{F})$ (Definition 29); the informal reading is that some positions or reference-classes are addressable under one candidate substrate and unaddressable under another with no canonical translation between them.

Remark 8. Pointing indeterminacy is intra-substrate: a single geometry with a single grammar leaves a gap. Substrate indeterminacy is inter-substrate: different geometries or grammars for the same statement produce non-comparable admissibility. Both modes are ISG-native; classical Boolean logic collapses both.

The two modes will drive the two flavours of DIA (Section 4) and, correspondingly, two flavours of REI; Figure 1 draws the two modes side by side.

3. Solid Integral, Atomic Geometry, and the SP–SR–IE Canon

3.1. **Two modes of specifying a spectrum.** Two contrasting presentation modes for spectra recur across the programme. We fix them as ISG-normative here.

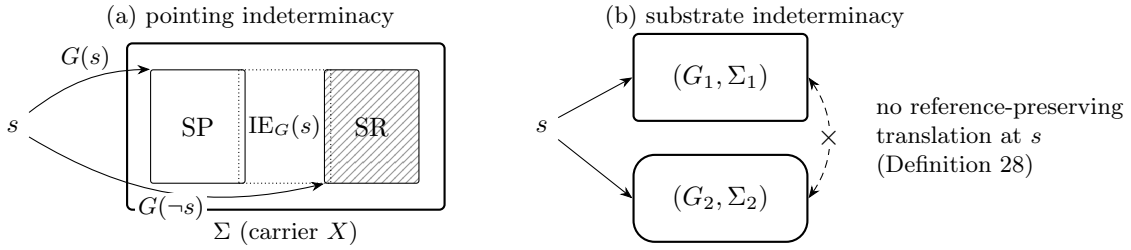


Figure 1: The two indeterminacy modes of Definitions 6 and 7. (a) Intra-substrate: a single grammar G on a single spectrum Σ addresses $\mathrm{SP}_G(s)$ and $\mathrm{SR}_G(s)$ and leaves the residual $\mathrm{IE}_G(s) = X \setminus (\mathrm{SP} \cup \mathrm{SR})$. (b) Inter-substrate: two candidate pairs both treat s with no reference-preserving translation between them (crossed arrow). The panels support the irreducibility claim of Section 2.4; they are not exclusive — a single statement can suffer both modes at once, the reflexive case of Section 6 being the paradigm. Region shapes and contiguity are drawn for legibility only; no topological claim is made here.

Definition 9 (Solid Integral mode)**.** A spectrum $\Sigma = (X, \mathsf{Adm}, \mathsf{Exc})$ is in *Solid Integral mode* when X carries a Borel measure or volume, Adm and Exc are generated by measurable primitive regions (typically metric balls or convex sets), and the reference of a grammar is set-membership against these primitive regions. The distinguishing feature: the substrate is given *extensionally*, as measurable material.

Definition 10 (Atomic Geometry mode)**.** A spectrum $\Sigma = (X, \mathsf{Adm}, \mathsf{Exc})$ is in *Atomic Geometry mode* when X is generated by atoms $\{\alpha_1, \alpha_2, \dots\}$ together with a set of combinatorial construction rules (intersection, union, complement, quantifier lift, product, projection). Adm and Exc are the images of admissible / inadmissible generation patterns. The distinguishing feature: the substrate is given *intensionally*, as a construction mechanism.

Example 11. The ADKL hypersphere realisation of a predicate space is Solid Integral: primitive regions are hyperspheres (c_i, r_i) in $\mathbb{R}^n$; admissibility is metric membership; the boundary zone $\partial_\epsilon A_i$ is a measurable diástēma [Bra26a]. The four-family $\mathsf{S/C/L/K}$ membership partition of Σ^ω-properties is Atomic Geometry: atoms are the four topological modalities, admissibility is the intersection lattice, the 16-cell partition is the generation [Bra26e]. The MCK region cube $\{0,1\}^3$ is also Atomic Geometry: atoms are the three predicates M, C, K; the seven regions R1–R7 are the admissible generation patterns [BAI26].

The two modes are not disjoint. Some substrates admit both presentations; the equivalence, when it exists, is a structural theorem-target of the framework.

Remark 12 (Solid $\leftrightarrow$ Atomic duality: an open direction). It is natural to ask when the Solid Integral and the Atomic Geometry presentation of a spectrum are inter-derivable. Two sufficient conditions can be sketched informally: a Solid-Integral Σ_{SI} with a finite partition of X into primitive regions induces an atomic presentation via the primitive regions as atoms and the region-lattice as the generation rules; an Atomic-Geometry Σ_{AG} with a finite atom set and a faithful representation into a measurable ambient space induces a Solid-Integral presentation via the measure of each admissible generation pattern. Neither direction is a theorem: both require non-trivial hypotheses (finiteness or, in the infinite case, descriptive-set-theoretic regularity of Adm; existence of a faithful measurable representation), and the

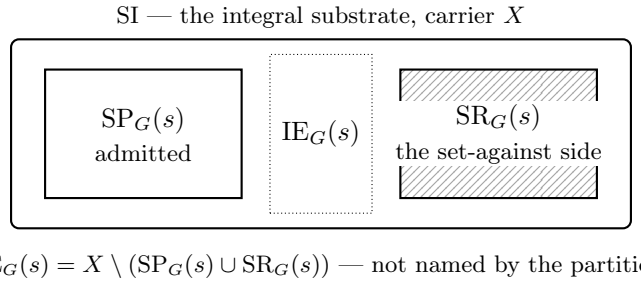


Figure 2: The SP–SR–IE canon (Section 3) in its historical two-solid reading: a partition act presents SP and SR as if jointly exhaustive; the dotted band is the excluded interval the presentation silently drops. The figure supports the convention above (DIA-freedom = empty IE) and nothing stronger: the band's position, size, connectedness, and geometry are illustrative — $\mathrm{IE}_G(s)$ is defined set-theoretically and need not be an interval of any order on X.

general correspondence over infinite or non-representable substrates is open. See Item (Q1) in Section 8.

3.2. **The SP–SR–IE canon.** Below the Solid/Atomic distinction sits a finer canon, historically introduced under Italian labels and here recovered under the canonical statement of [Bra26c]:

- SP (*Solido Proiettivo*, projective solid / admitted region): the distinguished subset of Σ that a grammar explicitly addresses under s. In the historical two-solid partition, SP is the reduced-dimensionality first solid that "restricts the critical dimension" of the original solid; here generalised to any admitted region. Formally $\mathrm{SP}_G(s) := \rho_G(s)$ when $G(s) \in \mathsf{Adm}$.
- SR (*Solido Residuale*, residual solid / forbidden region): the distinguished subset a grammar explicitly excludes. Historically, SR is the second solid of the partition, "presented as autonomous alternative", that gathers what the SP has left behind. Here generalised to any explicitly excluded region. Formally $\mathrm{SR}_G(s) := \rho_G(\neg s)$ when $\neg s$ is addressable and $G(\neg s) \in \mathsf{Exc}$. In the historical two-solid partition the residual side is itself an admitted region ($G(\neg s) \in \mathsf{Adm}$, the "autonomous alternative"); there the same formula $\mathrm{SR}_G(s) := \rho_G(\neg s)$ denotes that residual admitted side, and the explicitly-excluded case above is recovered when the alternative is named forbidden. In both readings SR is the side the partition sets against SP; the meaning of SR is unchanged.
- IE (*Intervallo Escluso*, excluded interval): $\mathrm{IE}_G(s) := X \setminus (\mathrm{SP}_G(s) \cup \mathrm{SR}_G(s))$. The region of meaning not covered by SP nor by SR, necessary for the continuity of the original substrate, and not named explicitly by the partition.

Convention. The SP–SR–IE trichotomy is the elementary DIA datum: DIA-freedom under (G, Σ) for s means $\mathrm{IE}_G(s) = \emptyset$. Every subsequent development of DIA (Section 4) either computes, bounds, or reintegrates IE. Figure 2 draws the datum on the historical two-solid reading.

Remark 13 (On terminology and historical grounding). *[METHODOLOGICAL / historical grounding.]* The Italian sub-canon SP–SR–IE, together with the SI–AG distinction (Definitions 9 and 10), originates in earlier programme notes recorded as internal conversations and never migrated to workspace documents. In those notes: (i) SI (*Solido Integrale*) is the original substrate, *continuous on all relevant dimensions*, with intermediate regions semantically valid; (ii) SP / SR is a two-solid partition of SI, with the specific rhetorical function of *making it appear that nothing has been left out*; (iii) IE is the non-addressed continuity that the partition silently drops, glossed with the technical synonyms *Vuoto di Continuità*, *Zona di Scomparsa Semantica*, or *Banda Cieca di Range*. The present paper preserves the abbreviations, generalises SP/SR from the specific two-solid partition to any admissibility partition, retains IE with its original meaning, and treats the historical two-solid case as the canonical two-atom Atomic-Geometry partition.

Remark 14 (The AS/AP/AR dialogic roles). *[HEURISTIC / programme-internal grounding for the SP/SR/IE terminology.]* The earlier notes also introduce three *dialogic roles* attached to a grammar-with-substrate: *Agente Sintetizzatore* (AS, synthesiser) responsible for maintaining the integral view of SI; *Agente Partizionatore* (AP, partitioner) responsible for producing the SP/SR split; *Agente Ricompositore* (AR, recomposer) responsible for detecting the excluded interval and running REI. In the reflexive ISG setting (Section 6), the same author, reader, or reviewer can occupy multiple roles across the same paper, and the paper's positioning discipline is a specific case of AP–AR interaction on the paper's own admissible-readings spectrum. We do not develop the AS/AP/AR calculus formally in this paper; it is retained here as programme-internal grounding for the SP/SR/IE terminology.

4. The Generative Cycle: from Constitutive DIA through Symbolic Genesis to REI Discipline

Sections 2–3 treated the spectrum $(\Sigma, \mathsf{Adm}, \mathsf{Exc})$ as a *given*: a substrate on which statements point via a grammar, and on which DIA/REI operate. This section unfolds the generative cycle that produces such a substrate in the first place. Three generative stages precede the static DIA/REI analysis on a given substrate: a *constitutive* partition act that generates the substrate together with its first excluded interval (Section 4.1); a *grammar-selection* meta-distinction that generates the capacity to refer to distinctions (Section 4.2); and a *symbolic-genesis* fixed-point that produces the symbolic content on which every downstream ISG operation lives (Section 4.3). The two indeterminacy modes named in Section 2.4 then instantiate as DIA (Sections 4.4 and 4.5); the REI discipline of Section 4.6 closes the cycle. Section 4.8 states the foundational theorem-target on the cycle's fixed point. Figure 3 draws the cycle with the four REI routes and the special standing of route (R3).

The generative cycle is not a novelty of the present paper's formalism; it is the recovered structure of two earlier programme observations — one cosmological, one MCK-side — that the framework was designed to formalise. Its cosmological grounding is recorded in Remark 16. Its MCK-side grounding lives in the pair {[BAI26],[Bra26b]}: M/C are established primitively in [BAI26], and K is generated by functional-closure iteration in [Bra26b] with an *inevitability* theorem at a specific *stabilization depth*. Under ISG, these two grounding lines converge: C is the product of the constitutive stage, M is the product of symbolic genesis, and K is the fixed point of iterated REI on the grammar-selection stage (Remark 20). The section here is the ISG-side formalisation of that convergence.

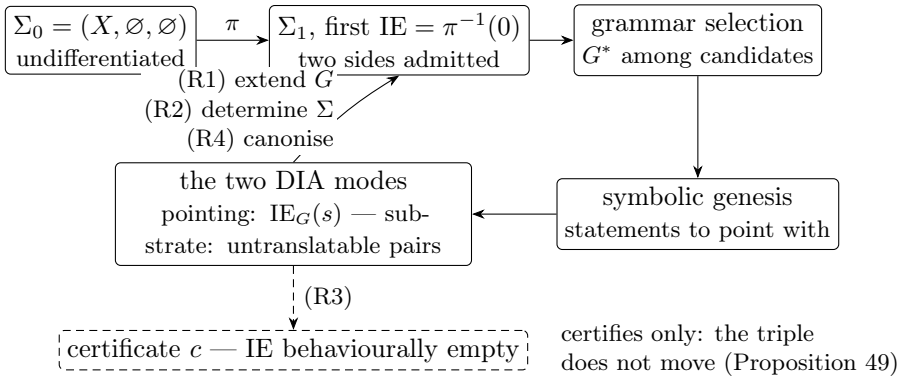


Figure 3: The generative cycle of Section 4: a constitutive partition act π produces the first substrate and its first excluded interval (Section 4.1); grammar selection and symbolic genesis produce the capacity to point (Sections 4.2 and 4.3); the two DIA modes then operate on the result, and the REI routes (R1), (R2), (R4) return to transform the state. Route (R3) is drawn dashed and apart because it is the one route that transforms nothing: it attaches a certificate and leaves the triple where it found it (Proposition 49). Arrows depict operational dependency, not implication; no claim about uniqueness or termination of the cycle is drawn.

4.1. **Constitutive DIA.** The primary DIA is not observational — naming a gap in a pre-existing substrate — but *constitutive*: the partition act generates the substrate and the excluded interval together. The archetype is cosmogonic: a primordial undifferentiated body undergoes a division act; the two resulting bodies and the between-them-space (the sky, the *διάστημα*) arise simultaneously (see Remark 16).

Definition 15 (Constitutive DIA)**.** Let $\Sigma_0 = (X, \varnothing, \varnothing)$ be an *undifferentiated substrate*: a support set with no admissibility structure yet declared. A *constitutive DIA event* at Σ_0 is a triple (π, G_1, Σ_1) where:

- $\pi : X \to \{+, -, 0\}$ is a *partition act* that assigns each point of X to one of two admitted sides ($+$ or $-$) or to the interval-between (0).
- G_1 is the induced grammar with domain the set of two labelling-statements $\{s_+, s_-\}$ and $G_1(s_+) := \pi^{-1}(+)$, $G_1(s_-) := \pi^{-1}(-)$.
- $\Sigma_1 = (X, \mathsf{Adm}_1, \mathsf{Exc}_1)$ is the induced spectrum with $\mathsf{Adm}_1 := \{\pi^{-1}(+), \pi^{-1}(-)\}$ and $\mathsf{Exc}_1 := \varnothing$; the interval-between $\pi^{-1}(0) = X \setminus (\mathsf{Adm}_1 \cup \mathsf{Exc}_1) = \mathrm{IE}_{G_1}(s_\pm)$ arises as the excluded interval induced by the act.

$\mathrm{DIA}_{\mathrm{const}}(\pi; \Sigma_0)$ denotes the excluded interval $\pi^{-1}(0) \subseteq X$. The constitutive DIA is *trivial* when $\pi^{-1}(0) = \varnothing$; otherwise the act produces a non-trivial diástēma whose subsequent addressability depends on the second stage.

Remark 16 (Cosmological grounding). *[HEURISTIC / etymological grounding.]* The Hesiodic and pre-Socratic cosmogonies encode the constitutive DIA as an origin myth. In Hesiod's *Theogony* (lines 116–138) [Hes66], the primordial *cháos* is differentiated by the separation of *Gaîa* and *Ourános*; the sky *ourános* is not a pre-existing body but the space that emerges from the separation act. In Anaximander (DK 12B1) [KRS83], the *ápeiron* (unbounded) separates into opposites through *diákrisis* (division). The Greek word *diástēma* ("interval, gap, the space-between") captures the interval produced by the separation act;

the term DIA of this framework is the direct etymological descendant. The framework's $\mathrm{DIA}_{\mathrm{const}}$ operator recovers this constitutive reading formally: the primary DIA produces the substrate and the diástēma together, rather than describing a gap in a substrate given elsewhere.

4.2. **Grammar-selection DIA (meta-distinction).** Once the substrate Σ_1 of a constitutive act is available, a new indeterminacy arises at a meta-level: the capacity to *distinguish the distinction* — to treat the partition itself as an object of admissibility rather than as a background structure. This meta-distinction operates not on Σ_1's points but on the space of grammars that could have partitioned Σ_0 into Σ_1.

Definition 17 (Grammar-selection DIA)**.** Let Σ_0 be an undifferentiated substrate and let $\mathcal{G}(\Sigma_0)$ denote the space of all constitutive partition-acts π over Σ_0. A *grammar-selection meta-substrate* $\Sigma_1^* = (\mathcal{G}(\Sigma_0), \mathsf{Adm}^*, \mathsf{Exc}^*)$ is a spectrum in the sense of Definition 2 whose points are partition-acts and whose admissibility structure encodes "which grammars are legitimate distinctions". The *grammar-selection DIA* of a statement t that names a partition-act $\pi_t \in \mathcal{G}(\Sigma_0)$ under a meta-grammar G^* is

$$\mathrm{DIA}_{\mathrm{gsel}}(t; G^*, \Sigma_1^*) \;:=\; \mathrm{IE}_{G^*}(t) \;=\; \mathcal{G}(\Sigma_0) \setminus \big(\mathrm{SP}_{G^*}(t) \cup \mathrm{SR}_{G^*}(t)\big) :$$

the set of partition-acts the meta-grammar leaves neither admitted nor explicitly excluded, i.e. the layer of *difference in the capacity to distinguish itself*.

Remark 18 (Thinness). *[HEURISTIC observation on the character of grammar-selection DIA.]* The grammar-selection DIA is a genuinely thin substrate: its points are not objects (celestial bodies, spectrum points, predicates) but partition acts. As earlier programme notes observe, its layer is "difficult to visualise" precisely because its operand is the act of drawing distinctions rather than any distinguished object. The pointing DIA (Definition 27) asks "where in Σ?"; the substrate DIA (Definition 29) asks "which Σ?"; the grammar-selection DIA asks "which act constitutes a Σ at all?".

4.3. **Symbolic genesis.** The grammar-selection stage has a distinguished effect that the pointing and substrate stages do not: it generates the capacity for *symbolic reference*. Where the constitutive stage yields distinguished objects (points of Σ_1) and the pointing stage locates statements on them, the grammar-selection stage yields the capacity to *refer* to a distinction as such — which is precisely what a symbol is.

Conjecture 19 (Symbolic genesis)**.** *Symbols do not pre-exist* $\mathrm{DIA}_{\mathrm{gsel}}$*. Specifically, let* Σ_{symb} *denote the space of intentional references (in the semiotic sense: signs that stand for something else, in a mind). Then* Σ_{symb} *arises as the fixed point of the iteration*

$$\Sigma^{(0)} := \Sigma_1 \quad \textit{(constitutive stage output)}, \qquad \Sigma^{(k+1)} := \Sigma^{(k)} \cup \mathcal{G}\big(\Sigma^{(k)}\big),$$

i.e. the closure of the substrate under iterated grammar-selection: at each step, the substrate is enriched by the space of partition-acts it currently supports. $\Sigma_{\mathrm{symb}} = \lim_{k\to\infty} \Sigma^{(k)}$ *when the iteration stabilises. The mode of stabilisation (finite* k*, transfinite, non-terminating) is a target for further work; the claim here is only that* Σ_{symb} *is a specific derivative of* $\mathrm{DIA}_{\mathrm{gsel}}$*, not a pre-existing background.*

Remark 20 (MCK genealogy). The generative cycle above gives the three MCK primitive predicates of [BAI26] a specific ISG-genealogy:

- C (*Contained*) = product of the constitutive stage (Definition 15): the delimited object arising from a partition act on an undifferentiated substrate.
- M (*Meaningful*) = product of symbolic genesis (Conjecture 19): a point of Σ_1 acquires M-status when the grammar-selection stage has made its distinction symbolic — i.e. addressable as a referent, not merely as an object. Here a "Σ_1-point" is a *domain element* (a point of the constitutive substrate), *not* an artifact already satisfying the C-predicate: M, C, K are three *independent* predicates on the same domain ([BAI26], Props. 2 and 4), so acquiring M neither presupposes nor entails C — cf. the non-empty pure-understanding cell $R_1 = \mathsf{M} \setminus (\mathsf{C} \cup \mathsf{K})$ of the region partition (*loc. cit.*, Theorem 1).
- K (*functional / generative*) = the fixed point of iterated REI (Section 4.6) on the grammar-selection DIA; equivalently, the point at which the meta-partitions stabilise. This is precisely the *stabilization depth* of [Bra26b], and the *inevitability theorem* there is, under ISG, the claim that this fixed point exists and is unique under a compositional-closure hypothesis.

We stress that MCK's primitive K is the *functionality* (operator-like, generative) predicate, *not* an epistemic "known" predicate: [BAI26] states explicitly that $\mathsf{K}(x) = 1$ "does not mean that x is 'known'" and that "K has functional character", reserving the epistemic notion for the *derived* set KNOWN $= \mathsf{M} \cup \mathsf{C}$. The ISG gloss above therefore names the *genesis stage* at which the functional capacity K first stabilises; it does not equate K with the derived epistemic set KNOWN. The convergence is not tautological; it decomposes into three distinct bridge obligations, each addressed by one of the lemmas below:

- *(T-C, category) — discharged.* The constitutive-DIA output is a partition region, whereas C in [BAI26] is a binary predicate on artifacts. Lemma 21 supplies the missing canonical predicate-from-region construction: C arises as the region-membership pullback $G^*[\mathsf{Adm}_1]$ along the denotation grammar (canonicity PROVEN), and it reproduces the MCK C predicate of [BAI26] under the constitutive-adequacy hypothesis (H-CA) (coincidence WITNESSED). The residue is the intensional/extensional discrepancy of Remark 22, not the category gap, which is closed.
- *(T-orth, orthogonality) — addressed as consistency.* [BAI26]'s cube $\{0,1\}^3$ requires $\mathsf{M}, \mathsf{C}, \mathsf{K}$ to be independent axes (*loc. cit.*, Props. 2 and 4), while the genealogy above is sequential (constitutive → symbolic → REI-fixed-point). Lemma 25 shows the sequential genesis is *consistent with* that orthogonality: the order constrains the definability of the three capacities, not their per-artifact values, and MCK's own independence results (Props. 2 and 4; Theorem 1) supply the seven non-empty cells as witnesses (consistency WITNESSED). What remains open (Remark 26) is the stronger claim that the genealogy *derives* the cell-structure rather than merely permitting it; that derivation is CONJECTURED.
- *(T-K, per-artifact vs global) — discharged.* The [Bra26b] stabilization depth is a global invariant of an architecture, whereas the operative K-status is per-artifact. Lemma 23 identifies the two: the global stabilization depth $D = n^*(\alpha)$ is the envelope $\sup_a d(a)$ of the per-artifact generation depths (identity PROVEN), and on the finite domain of the inevitability theorem $D < \infty$ forces uniform per-artifact K_α-membership (PROVEN under finite $\mathcal{A}$; the applicability of finiteness to the intended class is the WITNESSED

contingency). The identification holds on the finite-domain scope of [Bra26b]; the infinite-domain converse is the lemma's stated excluded interval (Remark 24).

With Lemmas 21, 23 and 25 all three tensions are addressed: (T-C) the region-to-predicate bridge is PROVEN as a canonical pullback, with the MCK coincidence WITNESSED under (H-CA); (T-K) the global–local depth identification is PROVEN on the finite-domain scope of [Bra26b]; and (T-orth) the genealogy is WITNESSED consistent with axis orthogonality. The remark is accordingly upgraded from schematic analogy to a *correspondence proved as consistency*; the derivation of MCK's cell-structure from the genealogy (Remark 26) remains open. The aftermath of the constitutive stage — the resulting (M_1, C, M_2) separation and the convergence of its middle residue with MCK's C — is developed in Section 4.9 (Definition 65).

Lemma 21 (Constitutive containment as a canonical region-pullback predicate)**.** *Let* $\Sigma_1 = (X, \mathsf{Adm}_1, \mathsf{Exc}_1)$ *be a constitutive spectrum (Definition 15) and let* $G : \mathcal{S} \rightharpoonup \mathsf{Adm} \cup \mathsf{Exc}$ *be a grammar (Definition 4) landing statements into regions of* X*. For any sub-family* $\mathcal{R} \subseteq \mathsf{Adm}$ *of admissible regions define the* region-membership pullback

$$G^*[\mathcal{R}](s) \;:=\; \big[\, s \in \mathrm{dom}(G) \text{ and } \rho_G(s) \in \mathcal{R} \,\big] \;\in\; \{0,1\}.$$

Then:

(i) (Canonicity — PROVEN.) *The assignment* $\mathcal{R} \mapsto G^*[\mathcal{R}]$ *is a Boolean-algebra homomorphism from* $(\mathcal{P}(\mathsf{Adm}), \cap, \cup, {}^{\mathrm{c}})$ *to the algebra of* $\{0,1\}$*-predicates on* $\mathrm{dom}_{\mathsf{Adm}}(G) := \rho_G^{-1}(\mathsf{Adm})$ *— the statements* G *lands in an admissible region, whose top element is* $\mathbf{1}_{\mathrm{dom}_{\mathsf{Adm}}(G)}$ *(statements landing in* Exc *map to* 0 *for every* $\mathcal{R}$*); it is the preimage (pullback) of region-membership along* G *and is the unique predicate-from-region assignment that commutes with* ρ_G*. Taking* $\mathcal{R} = \mathsf{Adm}_1$ *yields the* constitutive-containment predicate $\mathsf{C}_G := G^*[\mathsf{Adm}_1]$*:* $\mathsf{C}_G(s) = 1$ *iff* G *lands* s *in a delimited admitted region of the constitutive spectrum.*

(ii) (Coincidence with MCK C — WITNESSED, under (H-CA).) *Assume the* constitutive-adequacy hypothesis (H-CA)*: the artifact domain* $\mathcal{A}$ *of [BAI26] is denoted into* Σ_1 *by* G *so that* $\rho_G(a) \in \mathsf{Adm}_1$ *holds exactly for the artifacts that are epistemically present as delimited objects but not yet symbolically resolved. Then the extension of* C_G *equals the extension of the MCK predicate* C *(*Predicate C*, Definition 2 of [BAI26]).*

Proof. (i) Preimage under the partial function ρ_G preserves finite Boolean operations relative to $\mathrm{dom}_{\mathsf{Adm}}(G) = \rho_G^{-1}(\mathsf{Adm})$: for $\mathcal{R}, \mathcal{S} \subseteq \mathsf{Adm}$ and $s \in \mathrm{dom}_{\mathsf{Adm}}(G)$, $G^*[\mathcal{R} \cap \mathcal{S}](s) = G^*[\mathcal{R}](s) \wedge G^*[\mathcal{S}](s)$, and likewise for $\cup$ and for relative complement, since $\rho_G(s)$ is a single region tested for membership. Hence $\mathcal{R} \mapsto G^*[\mathcal{R}]$ is a Boolean homomorphism, namely the characteristic pullback ρ_G^{-1}. Uniqueness: any predicate-from-region assignment Φ satisfying $\Phi[\{R\}](s) = [\rho_G(s) = R]$ on singletons extends to at most one homomorphism, which is exactly the preimage. Instantiating $\mathcal{R} = \mathsf{Adm}_1$ gives C_G. *(ii)* By Definition 2 of [BAI26], $\mathsf{C}(x) = 1$ marks epistemic presence without semantic integration: x is registered (denoted) yet internally unresolved. By Definition 15, a region $R \in \mathsf{Adm}_1$ is a bare delimited admitted region produced by a partition act, prior to symbolic genesis (Conjecture 19) and to REI (Section 4.6) — i.e. a registered but not-yet-integrated object. Under (H-CA), $\rho_G(a) \in \mathsf{Adm}_1$ holds precisely for such artifacts, so C_G and MCK's C have equal extensions. The identification is grounded on the cited definition, not derived from region-membership alone (MCK's C is intensional); it is therefore witnessed under (H-CA), as recorded in Remark 22. □

Remark 22 (Residual content of (T-C)). *[Epistemic boundary of Lemma 21.]* Part (i) closes the *category* gap of (T-C) unconditionally: a region yields a predicate by a canonical functorial pullback. Part (ii) supplies something weaker than an identity of definitions. The MCK predicate C is *intensional* — $\mathsf{C}(x) = 1$ is the failure to construct a certificate of understanding ([BAI26], Definition 2) — whereas C_G is *extensional*, fixed by region-membership. Their coincidence is a grounded reading of the constitutive stage as "present-but-unresolved", contingent on (H-CA); it is WITNESSED, not PROVEN. Discharging the residue fully would require deriving certificate-failure from bare constitutive-region membership, which the present framework does not do.

Lemma 23 (Stabilization depth is the envelope of per-artifact generation depths)**.** *Let $\alpha = (\mathcal{O}, \circ)$ be a compositional architecture over a* finite *artifact domain $\mathcal{A}$, with generation sequence $(K_\alpha^{(n)})_{n\geq 0}$ and compositional closure $K_\alpha = \bigcup_{n\geq 0} K_\alpha^{(n)}$ ([Bra26b], Definition 2-F). For $a \in K_\alpha$ let $d(a) := \min\{\, n : a \in K_\alpha^{(n)} \,\}$ be the per-artifact generation depth (*loc. cit.*, Definition 3-F), with $d(a) := \infty$ for $a \notin K_\alpha$; set the per-artifact predicate $\mathsf{K}(a) := [\, d(a) < \infty \,] = [\, a \in K_\alpha \,]$. Let $D := n^*(\alpha) = \min\{\, n : K_\alpha^{(n+1)} = K_\alpha^{(n)} \,\}$ be the global stabilization depth (*loc. cit.*, Definition 5-F). Then:*

(i) (Envelope identity — PROVEN.) $D = \sup_{a\in K_\alpha} d(a) = \max_{a\in K_\alpha} d(a)$.

(ii) (Uniform K_α-membership — PROVEN under the finite-$\mathcal{A}$ hypothesis.) *$D < \infty$, and consequently $\mathsf{K}(a)$ holds — i.e. every $a \in K_\alpha$ is functionally reachable — under the single uniform bound $d(a) \leq D$. The finiteness $D < \infty$ is* deduced *in the proof from the finite-domain stabilization of [Bra26b] (Definition 2-F; and $n^*(\alpha) \leq |K_\alpha| - |\mathcal{O}|$ by Definition 5-F); the sole WITNESSED (modeling) contingency is whether the intended artifact class is finite, so that that hypothesis applies.*

(iii) (Global–local identification — PROVEN on finite $\mathcal{A}$.) *The per-artifact family $\{\mathsf{K}(a)\}_{a\in K_\alpha}$ and the global depth D determine one another: K is the indicator of $\{d < \infty\}$ and D is its envelope $\sup_a d(a)$.*

Proof. (i) The generation sequence is monotone non-decreasing ([Bra26b], Definition 2-F), so $K_\alpha^{(D)} = K_\alpha$ and every $a \in K_\alpha$ satisfies $d(a) \leq D$; hence $\sup_a d(a) \leq D$. If $D \geq 1$ and every a had $d(a) \leq D - 1$, then $K_\alpha^{(D-1)} = K_\alpha = K_\alpha^{(D)}$, i.e. $K_\alpha^{((D-1)+1)} = K_\alpha^{(D-1)}$, contradicting minimality of D; so some a has $d(a) = D$ and $\sup_a d(a) = D$. For $D = 0$, $K_\alpha = \mathcal{O}$ and all depths equal $0 = D$. Finiteness of $\mathcal{A}$ makes the supremum a maximum. *(ii)* By Definition 2-F, for finite $\mathcal{A}$ the sequence stabilizes in finitely many steps, and Definition 5-F bounds $n^*(\alpha) \leq |K_\alpha| - |\mathcal{O}| < \infty$; thus $D < \infty$. With (i), $d(a) \leq D < \infty$ for all $a \in K_\alpha$. *(iii)* Immediate from (i)–(ii): $\mathsf{K} = \mathbf{1}_{\{d<\infty\}}$ and $D = \sup_a d(a)$. □

Remark 24 (Excluded interval of Lemma 23). *[*DIA *acknowledgment.]* The envelope identity Lemma 23(i) holds as a supremum on any domain, but the uniform bound (ii) and the global–local identification (iii) use finiteness of $\mathcal{A}$. On an infinite artifact domain every per-artifact depth $d(a)$ may be finite while $D = \sup_a d(a) = \infty$: per-artifact K_α-membership then does *not* recover a finite global stabilization depth. The excluded interval is exactly this failure of uniformity; its REI requires a compactness or uniform-boundedness hypothesis, which the finite-domain inevitability theorem of [Bra26b] supplies precisely on finite $\mathcal{A}$.

Lemma 25 (Genealogical order is consistent with axis orthogonality)**.** *The sequential* ISG*-genealogy (constitutive → symbolic → REI-fixed-point) is logically consistent with the independence of the three MCK axes* M*,* C*,* K *([BAI26]). Precisely: the genesis order constrains*

only the definability order *of the three predicate-capacities at the framework level — the order in which the capacity to carry each predicate becomes available — and imposes no entailment among the per-artifact* value assignments $\mathsf{M}(x)$, $\mathsf{C}(x)$, $\mathsf{K}(x)$.

Proof. [Consistency implication PROVEN*; grounded on MCK's stipulated independence, so the composite statement is* WITNESSED*.]* A sequential availability order among capacities would induce an entailment among values only if a later capacity's activation were *necessary* for an earlier one, or conversely. No such necessity is present: [BAI26] *stipulates* the three predicates mutually independent — axis independence (*loc. cit.*, Proposition 2) and independence of K (*loc. cit.*, Proposition 4), both *definitional* commitments (posited, not derived) — so all eight triples $(\mathsf{M}, \mathsf{C}, \mathsf{K}) \in \{0,1\}^3$ are admissible. Under the standing seven-region model, in which all seven non-zero triples are realised as non-empty cells $R_1, \ldots, R_7$ (*loc. cit.*, Theorem 1 partitions them into disjoint, exhaustive regions; their individual non-emptiness is the model's admissibility assumption, not a consequence of the partition), the cells serve as value witnesses. In particular $R_1 = \mathsf{M} \setminus (\mathsf{C} \cup \mathsf{K})$, taken non-empty, witnesses $\mathsf{M} \not\Rightarrow \mathsf{C}$; $R_2 = \mathsf{C} \setminus (\mathsf{M} \cup \mathsf{K})$, taken non-empty, witnesses $\mathsf{C} \not\Rightarrow \mathsf{M}$; and the operator cells witness the corresponding non-entailments for K. A definability-order relation is compatible with every value triple — an available capacity may still take value 0 or 1 freely — so the genealogy's ordering of *when* each capacity becomes definable is consistent with the orthogonality of *what values* the predicates take. The consistency implication is elementary (PROVEN); it consumes MCK's independence results as inputs, which is why the composite claim is WITNESSED rather than self-contained. □

Remark 26 (Open residue of (T-orth): consistency vs derivation). *[CONJECTURED / explicitly open.]* Lemma 25 establishes that the genealogy is *consistent with* — i.e. does not contradict — the orthogonality of the MCK axes. It does *not* establish that the genealogy *derives* that orthogonality. The stronger claim (T-orth-derivation) — that the constitutive/symbolic/REI genealogy independently constructs a non-empty witness for each of the seven cells $R_1, \ldots, R_7$, so that the cell-structure is an *output* of the genesis rather than an externally supplied fact from [BAI26] — is left CONJECTURED. Discharging it would require exhibiting, from the genesis operators alone, an artifact realising each triple $(\mathsf{M}, \mathsf{C}, \mathsf{K})$; the present lemma borrows those witnesses from MCK's partition theorem instead of generating them.

4.4. **Pointing DIA.**

Definition 27 (Pointing DIA)**.** Let Σ be a spectrum and G a grammar. The *pointing DIA* of s under (G, Σ) is the excluded interval

$$\mathrm{DIA}_{\mathrm{point}}(s; G, \Sigma) \;:=\; \mathrm{IE}_G(s) \;=\; X \setminus (\mathrm{SP}_G(s) \cup \mathrm{SR}_G(s)).$$

The pair (G, Σ) is said to be DIA-*free at* s when $\mathrm{DIA}_{\mathrm{point}}(s; G, \Sigma) = \emptyset$.

Definition 27 recovers, in ISG notation, the excluded-interval calculus previously described informally at programme level. Concrete DIA scenarios — bounded model checking at depth k (excluded interval: traces of length $> k$); k-liveness (excluded interval: $k \to \infty$ residual); CEGAR (excluded interval: concrete refinement); Tseitin encoding (excluded interval: gate dependency structure) — are all instances of Definition 27 with specific choices of Σ and G.

4.5. **Substrate DIA.** The framework's genuinely new operator is the substrate-level one. Its definition needs care: for the intended cases, no canonical common ambient of the candidate substrates need exist, so symmetric-difference constructions on regions are not available. We instead cast the substrate DIA as an obstruction to reference-preserving translation across candidate (grammar, spectrum) pairs.

Definition 28 (Reference-preserving translation)**.** Let (G_i, Σ_i) and (G_j, Σ_j) be two (grammar, spectrum) pairs each treating a statement s, with $\Sigma_i = (X_i, \mathsf{Adm}_i, \mathsf{Exc}_i)$ and $\Sigma_j = (X_j, \mathsf{Adm}_j, \mathsf{Exc}_j)$. A *reference-preserving translation* from (G_i, Σ_i) to (G_j, Σ_j) at s is a partial function $\tau : X_i \rightharpoonup X_j$ such that

(i) τ is defined on all of $\mathrm{SP}_{G_i}(s) \cup \mathrm{SR}_{G_i}(s)$;
(ii) $\tau\big(\mathrm{SP}_{G_i}(s)\big) \subseteq \mathrm{SP}_{G_j}(s)$ and $\tau\big(\mathrm{SR}_{G_i}(s)\big) \subseteq \mathrm{SR}_{G_j}(s)$;
(iii) whenever X_i and X_j carry compatible ambient structure (topology, metric, order, or category), τ is a morphism of that structure on its domain.

Write $\mathcal{T}_{i\to j}(s)$ for the set of such translations; $\mathcal{T}_{i\to j}(s) = \emptyset$ is meaningful and asserts that no such τ exists.

Definition 29 (Substrate DIA)**.** Let s be a statement and $\mathcal{F} = \{(G_i, \Sigma_i)\}_{i\in I}$ a family of candidate (grammar, spectrum) pairs each treating s. The *substrate DIA* of s over $\mathcal{F}$ is the relation

$$\mathrm{DIA}_{\mathrm{sub}}(s; \mathcal{F}) \;:=\; \big\{\, (i,j) \in I \times I \;:\; \mathcal{T}_{i\to j}(s) = \emptyset \,\big\} \;\subseteq\; I \times I.$$

$\mathcal{F}$ is:

- *substrate-coherent at s* when $\mathrm{DIA}_{\mathrm{sub}}(s; \mathcal{F}) = \emptyset$;
- *substrate-preordered at s* when the complement $\{(i,j) : \mathcal{T}_{i\to j}(s) \neq \emptyset\}$ is a preorder on I (reflexive and transitive);
- *substrate-incoherent at s* otherwise.

Remark 30 (Well-definedness without common ambient)*.* $\mathrm{DIA}_{\mathrm{sub}}(s; \mathcal{F})$ is well-defined as a subset of $I \times I$ regardless of whether a canonical common ambient of the $\{X_i\}$ exists. Its cardinality and its structure (empty, preorder, incoherent) are cardinal invariants of the family $\mathcal{F}$ at s. When a canonical common ambient Y with embeddings $\iota_i : X_i \hookrightarrow Y$ does exist, the reference-preserving translations recover the classical symmetric-difference reading: $\mathcal{T}_{i\to j}(s) \neq \emptyset$ implies $\iota_j^{-1}\big(\iota_i(\mathrm{SP}_{G_i}(s))\big) \subseteq \mathrm{SP}_{G_j}(s)$ up to the embedding, so pointwise substrate disagreement is a special case of the translation obstruction. Substrate DIA is thus *always* a well-defined object; its finer combinatorial content (cocone structure, obstruction class in an appropriate cohomology) is a target for further work.

Remark 31. Substrate DIA deserves its own name because a preferred common substrate is often not available, and choosing one silently is an epistemic move. Definition 29 names this move by exhibiting the translation obstructions it would otherwise conceal.

4.6. **REI: four routes to reintegration.**

Definition 32 (REI)**.** A *reintegration of the excluded interval* of a pointing DIA at (G, Σ, s) is any one of the following operations that reduces or discharges $\mathrm{IE}_G(s)$ — "reduces" for the routes that shrink it as a set, "discharges" for (R3), which shows it behaviourally empty while leaving it as a set intact (Proposition 49):

(R1) **Grammar extension.** Replace G by an extension $G' \supseteq G$ that names some of $\mathrm{IE}_G(s)$ as SP or SR.
(R2) **Substrate determination.** Replace Σ by a refinement $\Sigma' \supseteq \Sigma$ that exposes further admissibility structure, allowing the unchanged G to name additional regions.
(R3) **Finite-confirmation certificate.** Supply a finite object (a Hansel–Newton tuple (D, A_0, U, E) [Bra26d], an inductive invariant, a k-induction witness) whose existence *implies* that $\mathrm{IE}_G(s)$ is behaviourally empty even if not syntactically so.

For a substrate DIA, a REI additionally admits:

(R4) **Canonisation.** Exhibit a canonical common embedding of the candidate substrates into a shared ambient, reducing the substrate discrepancy to a pointing one.

Remark 33. The three pointing routes correspond, in classical verification, to ((R1)) enriching the specification language; ((R2)) refining an abstraction (CEGAR-style); ((R3)) supplying an inductive invariant or k-induction step [SSS00; Bra11]. Canonisation ((R4)) has no classical counterpart: it is the ISG-native operation of naming a common substrate under which two disagreeing grammars can be compared.

Remark 34 (Atomic operators inside a REI act). *[HEURISTIC / practical decomposition; no formal operator calculus stated.]* Earlier programme notes decompose a single REI act into three atomic operators, applied in order regardless of which of R1–R4 delivers the reintegration: (i) *RNC* (*Rilevazione di Non-Contiguità*) — detection that the current partition $\mathrm{SP} \cup \mathrm{SR}$ leaves a non-trivial IE; (ii) *EIE* (*Esplicitazione dell'IE*) — naming the excluded interval as an explicit object rather than allowing it to remain silent; (iii) *RG* (*Ricomposizione a Gradiente*) — graded reintegration of the excluded content into the admissibility structure. The distinction between an R1–R4 route (mechanism used) and an RNC–EIE–RG operator sequence (steps taken) is orthogonal: any of the *state-transforming* routes $\{(R1), (R2), (R4)\}$ may be executed by exactly the same three-operator sequence. Route (R3) is excluded from this claim: its RG step would have to reintegrate content into the admissibility structure, and by Proposition 49 that route alters no component of the state. And any of the three operators may fail (RNC fails ⇒ no REI possible; EIE fails ⇒ REI without acknowledgment, which imports a fresh DIA; RG fails ⇒ non-graded reintegration, which typically corresponds to a bare canonisation of the substrate). We do not develop this operator calculus formally here; it is retained as a decomposition handle for practical use.

Remark 35 (The Hansel–Newton certificate for route R3). *[WITNESSED, grounded on the cited Hansel–Newton documentation.]* For (R3), the canonical certificate structure is the Hansel–Newton refinement tuple of [Bra26d], which we restate here (adapted) as it directly instantiates the REI discipline for infinite-trace properties. A *refinement system* for a target property $P \subseteq \Sigma^\omega$ is a tuple $R = (D, A_0, U, E)$ where

- $D = \{D_k\}_{k \in \mathbb{N}}$ is a sequence of *refinable domains*; each D_k is a set of hypotheses, candidates, or representations whose internal resolution increases with k.
- $A_0 \in D_1$ is a *coarse initial approximation* — a starting hypothesis or default state, guaranteed to be in the controlled initial domain.
- $U : D_k \to D_{k+1}$ is a *computable update rule*, refining a candidate from level k to a candidate at level $k+1$.
- $E : D_k \to \mathbb{N}$ (or $\mathbb{R}_{\geq 0}$) is a *computable error metric*: $E(A)$ measures the distance between a candidate $A \in D_k$ and the target property P.

The system R *certifies* P *at level* k when $E\big(U^k(A_0)\big) \leq \varepsilon$ for a chosen tolerance ε. The intended convergence criterion is $E(A_k) \to 0$ as $k \to \infty$, or $\exists k$ such that $E(A_k) \leq \varepsilon$. Under ISG, the four components admit a natural property-class encoding ([Bra26d], adapted): D corresponds to the Liveness dimension of P; A_0 to its Safety dimension; U to a combination of Enforceability and Constructive-Liveness; E to a combination of Monitorability and Cosafety. A route-R3 REI act on $\mathrm{IE}_G(s)$ thus consists in exhibiting a specific (D, A_0, U, E) that certifies the content of s that $\mathrm{SP}_G(s)$ does not syntactically deliver.

4.7. **A typed calculus for DIA and REI.** Sections 4.1 to 4.6 deliver four DIA operators and four REI routes, but not the calculus that governs them: which operator may be applied to what, which route may follow which operator, and what — if anything — the resulting act is worth. This subsection supplies that calculus. It discharges the account deferred in Remark 69 and fixes three distinctions that are routinely collapsed in informal use: the *type* of an operation, the *direction* in which it is applied, and its *normative valence*.

4.7.1. *Three axes.* The admissibility family Adm of a spectrum is typically not given by enumeration but generated by constraints. Naming that presentation is what makes the second axis expressible.

Definition 36 (Constraint presentation)**.** *[DEFINITIONAL.]* A *constraint presentation* of a spectrum $\Sigma = (X, \mathsf{Adm}, \mathsf{Exc})$ is a set Γ of constraints on subsets of X, written in a language L carrying a consequence relation $\vdash$, together with an *intersective* semantics

$$\mathrm{Mod}_X(\Delta) \;=\; \{\, S \subseteq X \;:\; S \models \gamma \text{ for every } \gamma \in \Delta \,\},$$

such that $\mathsf{Adm} = \mathrm{Mod}_X(\Gamma)$. Intersectivity is what makes Mod_X anti-monotone (Remark 37) and is assumed throughout. We require L to contain $\bot$ and $\neg$, and $\vdash$ to be a standard consequence relation over L: reflexive ($\gamma \in \Gamma \Rightarrow \Gamma \vdash \gamma$), monotone, transitive (closed under cut), closed under reductio ($\Gamma \cup \{\varphi\} \vdash \bot \Rightarrow \Gamma \vdash \neg\varphi$), and *sound* for Mod_X — if $\Gamma \vdash \varphi$ then every member of $\mathrm{Mod}_X(\Gamma)$ satisfies φ. Where L carries modal operators $\Box, \Diamond$ we assume its modal fragment is normal, so that $\neg\Diamond\varphi$ and $\Box\neg\varphi$ are interderivable. Constraints hold of *members* of Mod_X, that is of subsets of X; we write them accordingly. A spectrum may admit many presentations, or none that is finitely given; the calculus below is stated relative to a chosen Γ and is silent where no presentation is fixed.

Remark 37 (Anti-monotonicity of Γ). *[PROVEN; immediate.]* Mod_X is anti-monotone: $\Gamma' \subseteq \Gamma$ implies $\mathrm{Mod}_X(\Gamma) \subseteq \mathrm{Mod}_X(\Gamma')$. Removing a constraint therefore *enlarges* the admissible region. This is the formal content of the "over-constrained target" case of Remark 69.

Definition 38 (Direction of application)**.** *[DEFINITIONAL.]* Let $\Sigma = (X, \mathsf{Adm}, \mathsf{Exc})$ carry a constraint presentation Γ. An operation is applied

- *object-directed* when its target is X — it partitions or restricts the carrier (or a member of Adm), leaving Γ fixed;
- *constraint-directed* when its target is Γ, leaving X fixed;
- *reflexive* when its target is the question whether a given $\gamma \in \Gamma$ belongs to the presentation at all.

Direction names the *target* of the act, not the polarity of the change it produces. The two must be kept apart, because a constraint-directed act can move the admissible region either way: by Remark 37 a *subtractive* act, removing some γ, does not shrink Adm and typically enlarges it, whereas an *additive* act, adding a constraint — as R1-aug does, Definition 48 — does not enlarge Adm and typically shrinks it. Where the polarity matters we say subtractive or additive explicitly; the liberatory reading of a cut (Remark 69) is the subtractive constraint-directed case specifically, not the constraint-directed case as such. Direction is defined relative to a *single* pair (X, Γ). For an operation ranging over a family of spectra — in particular for a substrate DIA (Definition 29), whose members need not share an ambient (Remark 30) — direction is read per member of the family, relative to that member's own presentation.

Definition 39 (Valence)**.** *[DEFINITIONAL.]* The *valence* of an application is a partial function

$$\nu \;:\; (\text{operation},\, \Sigma,\, \Gamma,\, a,\, g) \;\longrightarrow\; \{\text{beneficial, harmful, mixed, neutral}\},$$

partial because G and L below need not be evaluable — where either is undecided, ν is simply undefined at that argument, which is a different situation from the determinate no-gain-no-loss case that (V4) names. evaluated relative to an affected agent a and a goal g, and constrained by the following *adequacy clauses*. Write $(X, \Gamma) \rightsquigarrow (X', \Gamma')$ for the application, and compare the admissible fields $\mathrm{Mod}_X(\Gamma)$ before and $\mathrm{Mod}_{X'}(\Gamma')$ after. The comparison is semantic, so that an application which moves the carrier and one which moves the constraints are both evaluable. Write G for the *gain* condition — g is realisable in $\mathrm{Mod}_{X'}(\Gamma')$ and not in $\mathrm{Mod}_X(\Gamma)$ — and L for the *loss* condition: either g is realisable in $\mathrm{Mod}_X(\Gamma)$ and not in $\mathrm{Mod}_{X'}(\Gamma')$, or some φ on which g depends holds of every member of $\mathrm{Mod}_X(\Gamma)$ and fails of some member of $\mathrm{Mod}_{X'}(\Gamma')$. The four clauses are mutually exclusive and exhaustive:

(V1) $\mathsf{G} \wedge \neg\mathsf{L} \;\Rightarrow\; \nu = \text{beneficial}$;
(V2) $\mathsf{L} \wedge \neg\mathsf{G} \;\Rightarrow\; \nu = \text{harmful}$;
(V3) $\mathsf{G} \wedge \mathsf{L} \;\Rightarrow\; \nu = \text{mixed}$;
(V4) $\neg\mathsf{G} \wedge \neg\mathsf{L} \;\Rightarrow\; \nu = \text{neutral}$.

By soundness, $\Gamma \vdash \varphi$ is a sufficient syntactic condition for φ to hold of every member of $\mathrm{Mod}_X(\Gamma)$, which is how (V2) is checked in practice on the constraint-directed side. Valence is relative to (a, g) and is not a property of the operation. Remark 69 reads the same relativity against a coherence/freedom baseline: (V1) covers the freedom-restoring cut on an over-constrained target, and (V2) the fragmenting cut on a coherent whole — the latter typically through its first disjunct, since fragmenting the carrier is what costs an agent a realisation.

By an *application* we mean an operator of Sections 4.1 to 4.5 together with the act it licenses on (X, Γ); direction is a property of that act, not of the operator taken alone.

Proposition 40 (The three axes are independent)**.** [PROVEN; (ii) under the stated hypothesis, (iii) a scope statement about genesis.]

(i) Direction does not determine valence. *There are two triples (application, agent, goal) sharing the same application — hence the same operator type and the same direction — whose valences differ.*
(ii) Operator type does not determine direction. *Let the operator be pointing or grammar-selection, and suppose the presentation* Γ *contains a constraint whose removal admits*

a region the operator has just exhibited as undecided. Then both the object-directed and the constraint-directed act are available on the same operator output. For a substrate operator the same holds per member of the family (Definition 38).

(iii) Genesis scope. *For a constitutive operator applied at a stage at which no constraint presentation is given, the constraint-directed reading is undefined; only the object-directed one is available.*

Proof. (i): we exhibit the application and the two triples. Let $X = \{x_1, x_2\}$ and let γ be the constraint holding of a subset $S \subseteq X$ exactly when $x_2 \notin S$. Put $\Gamma = \{\gamma\}$, so $\mathrm{Mod}_X(\Gamma) = \{\varnothing, \{x_1\}\}$. Let the application be the constraint-directed removal of γ, giving $\Gamma' = \varnothing$, $X' = X$ and $\mathrm{Mod}_X(\Gamma') = \mathcal{P}(X)$. Take a_1 with goal g_1 = "occupy an admissible set containing x_2": realisable after, not before, and no guarantee g_1 depends on is lost, so ν = beneficial by (V1). Take a_2 with goal g_2 depending on γ itself: γ holds of every member of $\mathrm{Mod}_X(\Gamma)$ and fails of $\{x_1, x_2\} \in \mathrm{Mod}_X(\Gamma')$, so L holds; and g_2 was already realisable before the application, since γ holds throughout $\mathrm{Mod}_X(\Gamma)$, so G fails. Hence ν = harmful by (V2). (In particular γ is not redundant: by reflexivity $\Gamma \vdash \gamma$, while $\Gamma' \nvdash \gamma$ by soundness, since $\{x_1, x_2\}$ falsifies it.) The two triples share one application and differ on ν.

(ii): for pointing, fix (Σ, G, s) with $\mathrm{IE}_G(s) \neq \varnothing$ and let $\gamma \in \Gamma$ be as hypothesised, so that some member of $\mathrm{Mod}_X(\Gamma \setminus \{\gamma\})$ meeting $\mathrm{IE}_G(s)$ is *newly* admitted, i.e. lies outside $\mathrm{Mod}_X(\Gamma)$. The object-directed act restricts attention to $X \setminus \mathrm{IE}_G(s)$, leaving Γ fixed; the constraint-directed act removes γ, leaving X fixed and, by Remark 37, enlarging the admissible field. Both are acts on (X, Γ) licensed by the same computation of $\mathrm{IE}_G(s)$, so the operator does not determine the direction. For grammar-selection (Definition 17) the meta-substrate is $\mathcal{G}(\Sigma_0)$ and the hypothesis reads: some adequacy requirement, on the candidate grammars, whose removal readmits a grammar the operator has just exhibited as undecided between. The object-directed act discards a candidate grammar from the carrier; the constraint-directed act discards that requirement. For a substrate operator, apply either act inside a fixed member of the family, as Definition 38 directs; the act is then the corresponding pointing act on that member, which is what per-member reading means.

We record that the hypothesis in (ii) is a genuine restriction and not an idle one: $\mathrm{IE}_G(s)$ is defined by failure of reference (Section 3), not by inadmissibility, so nothing in the framework guarantees that any single $\gamma \in \Gamma$ is responsible for it.

(iii): Definition 15 posits Σ_1 from an undifferentiated Σ_0 by a partition π. At that stage no Γ with $\mathsf{Adm}_1 = \mathrm{Mod}_X(\Gamma)$ has been *given*, so Definition 38, which is stated relative to a chosen Γ, does not define the constraint-directed reading. (Once any presentation is fixed — and one membership constraint presents Adm_1 — the restriction lapses. The claim is about what is available at the constitutive stage, not about what could later be supplied.) □

Remark 41 (Adapter-induced restrictions). *[DEFINITIONAL bookkeeping; see Table 2.]* Proposition 40(iii) is not the only place where an axis constrains another. Two adapters of Table 2 carry the same effect and are recorded here rather than left implicit: A3 and A5 are genesis companions to (iii) and likewise admit only the object-directed reading, and A4 forces the constraint-directed reading, since a pair-lifted pointing (Definition 47) interrogates a constraint that presupposed common measurability rather than a single statement's reference.

Remark 42 (Possibility reopening). *[PROVEN; elementary, and load-bearing.]* Suppose $\Gamma \vdash \Box\neg F$ and a constraint-directed application removes a γ such that $\Gamma' := \Gamma \setminus \{\gamma\} \nvdash \Box\neg F$. (The weaker hypothesis "γ occurs in some derivation of $\Box\neg F$" does not suffice: another derivation

may avoid γ.) Since the modal fragment of L is normal (Definition 36), $\neg\Diamond F$ and $\Box\neg F$ are interderivable, so $\Gamma' \nvdash \neg\Diamond F$. By closure under reductio (same definition), contraposed, this gives $\Gamma' \cup \{\Diamond F\} \nvdash \bot$: the reopened possibility is consistent with what survives the cut. It does *not* follow that F, nor even that $\Gamma' \vdash \Diamond F$. Cutting the grounds of an unfavourable necessity establishes at most the reopening of a favourable possibility.

4.7.2. *Separators.* A DIA operator says *how* a distinction is installed; it does not say *on what criterion*. The criterion is a separate typed datum.

Definition 43 (Separator)**.** *[DEFINITIONAL.]* A *separator* is a tuple $\mathsf{S}_\Phi = (B, \Phi, W, V, \Pi)$ where B is the body separated (a spectrum, a model class, a language, a family of maps); Φ is the separating criterion, a typed predicate on B; $W = W^+ \sqcup W^-$ is a witness space, its two parts carrying witnesses for and against Φ; V is a verifier, in general partial, which on input $b \in B$ searches W and returns $+$ on producing a member of W^+ for b, $-$ on producing a member of W^-, and nothing otherwise; and $\Pi = (B^+_\Phi, B^-_\Phi, B^?_\Phi)$ is the partition of B induced by those three outcomes.

Remark 44 (Reading a separator as a pointing)*. [DEFINITIONAL; a stipulated reading, not a theorem.]* We read the third block $B^?_\Phi$ — the sub-body on which V returns no verdict — as an excluded interval in the sense of the SP–SR–IE canon (Section 3), taking B as carrier, B^+_Φ as SP and B^-_Φ as SR. This is a *convention* fixing how a criterion is presented to the REI apparatus, not a theorem: we do not claim that every separator arises as ρ_G for some antecedently given grammar. Under the convention, $B^?_\Phi$ is the object on which the routes of Definition 32 act, which is what Table 3 presupposes.

Remark 45 (Naming)*. [Naming note; no formal content.]* Internal programme notes call a separator a *sword of* DIA. The metaphor is expository: the tuple of Definition 43 carries the formal weight, and we use "separator" throughout.

Table 1 lists the criteria the programme has needed so far. The list is a working catalogue, not a claim of exhaustiveness.

4.7.3. *Adapters.* Several criteria of Table 1 are not directly applicable under every DIA operator: a criterion may presuppose structure the operator does not supply. Table 2 names what must be supplied in each such case. Two of the adapters need their mechanics fixed.

Definition 46 (Atomic stipulation)**.** *[DEFINITIONAL.]* A constitutive DIA posits (Σ_1, σ) *atomically* when the substrate and the structural datum σ (an equivalence, an order, a measure, a role alphabet, an operator signature) are fixed by one stipulation, with no intermediate stage at which Σ_1 stands and σ does not. The device is needed because a criterion such as Incongruence has nothing to fail against on a substrate posited without a $\sim$, while a sequential stipulation would require $\sim$ to be available as an object before Σ_1 admits equivalence-typing.

Definition 47 (Pair-lifted pointing)**.** *[DEFINITIONAL.]* The *pair lift* of a pointing DIA lifts the admitted and excluded regions along the product,

$$\mathrm{SP}^*_G(s_1, s_2) := \mathrm{SP}_G(s_1) \times \mathrm{SP}_G(s_2), \qquad \mathrm{SR}^*_G(s_1, s_2) := \mathrm{SR}_G(s_1) \times \mathrm{SR}_G(s_2),$$

Separator	Criterion Φ	Type of criterion
Falsity	x admits a countermodel	model-theoretic
Incompleteness	$\exists y.\, x \nvdash y \wedge x \nvdash \neg y$	proof-theoretic
Inaccuracy	$d(x, y^*) > \varepsilon$	metric
Incomputability	no decision procedure for x	recursion-theoretic
Incongruence	f does not preserve $\sim$	structural
Incomparability	$a \not\preceq b \wedge b \not\preceq a$	order-theoretic
Incommensurability	no common measure for A_1, A_2	measure-theoretic
Incoherence	Γ generates incompatible roles	grammar-side
Inconsistency	$\Gamma \vdash \bot$	proof-theoretic
Infinity	x is infinite (cardinality; or non-terminating)	cardinality / safety
Non-linearity	f violates superposition	structural

Table 1: Separating criteria used in the programme (Definition 43). Incomparability and Incommensurability are kept apart deliberately: the absence of an order is weaker than the absence of a common measure. Incoherence and Inconsistency are likewise distinct: the first is a condition on what a grammar generates, the second a condition on what a constraint set proves. The Infinity row carries two readings that the four-family classification keeps apart: a cardinality reading, which asserts nothing behavioural, and a non-termination reading, which is. The second is $\mathsf{G}\neg \mathit{halt}$: every violating run has a finite bad prefix, namely the one ending at the halt event, so non-termination is a *safety* property, and its complement — termination, $\mathsf{F}\, \mathit{halt}$ — is the cosafety side, finitely confirmable on a good prefix [Bra26e]. Route (R3) acts on the second reading, and the Hansel–Newton encoding of Remark 35 spans all four families rather than singling one out.

and takes as excluded interval of the pair everything jointly undecided,

$$\mathrm{IE}^*_G(s_1, s_2) \;:=\; (X \times X) \setminus \big(\mathrm{SP}^*_G(s_1, s_2) \cup \mathrm{SR}^*_G(s_1, s_2)\big).$$

A pair thus falls in IE^* unless both coordinates are admitted or both are excluded — which is the reading a criterion comparing two statements requires. The lift does *not* recover the unlifted operator wholesale: $\mathrm{IE}^*_G(s, s)$ contains every mixed pair (x_1, x_2) with $x_1 \in \mathrm{SP}_G(s)$ and $x_2 \in \mathrm{SR}_G(s)$, and such pairs have no counterpart in $\mathrm{IE}_G(s)$. The recovery holds on the diagonal $\Delta = \{(x, x) : x \in X\}$ only:

$$\Delta \cap \mathrm{IE}^*_G(s, s) \;=\; \{(x, x) : x \notin \mathrm{SP}_G(s) \text{ and } x \notin \mathrm{SR}_G(s)\} \;\cong\; \mathrm{IE}_G(s).$$

Other liftings (diagonal-only, or via a primitive joint-interpretation predicate) are possible and are not used here.

4.7.4. *Two sub-routes of R1, and the standing of R3.*

Definition 48 (Sub-routes of grammar extension)**.** *[DEFINITIONAL; a refinement of (R1), not a redefinition.]* Route (R1) splits according to whether the constraint presentation moves with the grammar:

- *R1-pure*: $G \rightsquigarrow G'$ only, with Γ unchanged;
- *R1-aug*: $G \rightsquigarrow G'$ together with a scoped augmentation of Γ by a replacement constraint γ' expressible in G'.

	What must be supplied
A1	A translation-invariant Mod or $\vdash$, so that a countermodel or a proof of $\bot$ is recognisable across the members of a substrate family. Concretely: bi-interpretability, a reflection principle over a shared object language, or a semantic embedding between the two universes.
A2	A shared object language across the candidate grammars of a grammar-selection meta-family, so that the criterion is meaningful in every member — a common arithmetic fragment, or a common role alphabet.
A3	A relational datum ($\sim$, $\preceq$, a metric, or a measure) fixed at genesis by atomic stipulation (Definition 46), for the structural and measure-theoretic criteria.
A4	A pair-lifted pointing (Definition 47), for criteria that compare two statements rather than evaluate one.
A5	A grammar signature or operator signature fixed at genesis by atomic stipulation, for grammar-side and structural criteria. A5 does *not* supply a constraint presentation: criteria defined on Γ remain unavailable at genesis until Γ is itself stipulated.

Table 2: Adapters. An adapter is not a repair: it is the structure a criterion presupposes, without which the corresponding cell of Table 3 is not defined rather than false.

R1-aug is the constraint-directed sub-route. A bare exchange of constraints with no accompanying grammar extension is not an instance of (R1) at all; it requires a canonisation ((R4)) of the revised presentation. Notation: a cross-reference to (R1) denotes the route as a whole, as defined in Definition 32; the names "R1-pure" and "R1-aug" are used whenever the sub-route matters, and are always written out.

Proposition 49 (Route R3 certifies and does not transform)**.** [PROVEN.] *Let a route-(R3) act at (Σ, G, s) supply a certificate c. Then its output type is the pair*

$$\big((\Sigma, G, \Gamma),\ c\big),$$

whose first component is the input triple unchanged. Writing π_1 for the first projection, $\pi_1 \circ \mathrm{REI}_3 = \mathrm{id}$ on the triples; hence for any operation D with domain the triples, $D \circ \pi_1 \circ \mathrm{REI}_3 = D$. Route (R3) contributes no state change to any composite in which it occurs.

Proof. By Definition 32(R3) the act supplies a finite object whose existence implies that $\mathrm{IE}_G(s)$ is behaviourally empty. The clause specifies the supply of that object and nothing further; in particular it fixes no new value for any component of (Σ, G, Γ), so $\pi_1 \circ \mathrm{REI}_3$ is the identity on the triples and $D \circ \pi_1 \circ \mathrm{REI}_3 = D$. (For orientation, the contrast with the other three routes: (R1) names elements of $\mathrm{IE}_G(s)$ as SP or SR, (R2) replaces Σ, and (R4) exhibits an embedding of substrates; each therefore does fix a new value.) What the act changes is the epistemic standing of $\mathrm{IE}_G(s)$, which is recorded in the second component and is not a component of the triple. □

Remark 50 (The R3 chain). *[PROVEN corollary of Proposition 49; discipline statement.]* A state change grounded on a route-R3 certificate must therefore be written with its intermediate stages visible:

$$(\Sigma, G, \Gamma) \xrightarrow{\text{REI}_3} \big((\Sigma, G, \Gamma),\, c\big) \longrightarrow \text{admissible decision procedure} \longrightarrow \text{action} \longrightarrow (\Sigma', G', \Gamma').$$

Only the labelled arrow is a REI act in the sense of Definition 32, and by Proposition 49 it leaves the triple where it found it; the three unlabelled arrows are outside the framework, and they are where the state actually moves. Compressing the chain — writing REI_3 as though it delivered (Σ', G', Γ') — is the error this remark exists to block. In particular $\{(R1), (R2), (R4)\}$ are the state-transforming routes, and (R3) is not among them.

The next result is stated for the constitutive operator only. The other three DIA operators of Sections 4.2 to 4.5 compute a set or a relation and do not by themselves move (Σ, G, Γ); for them the question of inversion does not arise in the same form, and asking a REI route to invert them would be a category confusion rather than a false claim.

Proposition 51 (Constitutive DIA is not invertible)**.** [PROVEN.] *Let* (π, G_1, Σ_1) *be a constitutive* DIA *event at* $\Sigma_0 = (X, \varnothing, \varnothing)$ *(Definition 15), and suppose* $\pi^{-1}(+)$ *and* $\pi^{-1}(-)$ *are not both empty. Then no composite of* REI *route applications carries* Σ_1 *back to* Σ_0*.*

Proof. Say a spectrum is *inhabited* when $\mathsf{Adm} \cup \mathsf{Exc} \neq \varnothing$. By Definition 15, $\mathsf{Adm}_1 = \{\pi^{-1}(+), \pi^{-1}(-)\}$, so Σ_1 is inhabited under the hypothesis, while Σ_0 is not. It therefore suffices to show that each route preserves inhabitation; the claim then follows by induction on the length of the composite.

(R1): the route requires G' to name some element of $\text{IE}_G(s)$ as SP or SR, a condition evaluated at the post-state (G', Γ') — which matters here, since R1-aug moves grammar and presentation together. By the canon (Section 3) a statement is named SP or SR only when its reference lies in $\mathsf{Adm}' \cup \mathsf{Exc}'$; so a successful application leaves $\mathsf{Adm}' \cup \mathsf{Exc}'$ inhabited. This covers both sub-routes of Definition 48. It is worth recording what the requirement excludes, since it is the one near-miss: an R1-aug act that adjoins an unsatisfiable γ' gives $\text{Mod}_X(\Gamma') = \varnothing$ and hence $\mathsf{Adm}' = \varnothing$, which together with $\mathsf{Exc}_1 = \varnothing$ would be Σ_0 exactly. Such an act is not an instance of (R1): with $\mathsf{Adm}' \cup \mathsf{Exc}' = \varnothing$ nothing can be named SP or SR, so the route's own defining requirement fails.

(R2): the route replaces Σ by a refinement $\Sigma' \supseteq \Sigma$, so $\mathsf{Adm}' \supseteq \mathsf{Adm}$ and inhabitation is preserved.

(R3): by Proposition 49 the triple is unchanged, so inhabitation is preserved trivially.

(R4): the route exhibits a common ambient together with embeddings ι_i of the candidate substrates. Each ι_i is reference-preserving in the sense of Definition 28, whose clause on the admitted side carries SP into SP; the image of an inhabited substrate is therefore inhabited, and so is the ambient. □

Remark 52 (What the proposition does and does not say). *[DEFINITIONAL; scope statement.]* Proposition 51 is a statement about *recovery of a state*, not about the four routes being non-invertible as maps. It says that the particular state Σ_0 is unreachable from Σ_1, and it says so by an invariant that no route can break. Two consequences worth separating. First, reintegration is not restoration: a REI act following a constitutive cut moves the state forward, not back, and Remark 68's suture — a canonisation that merges M_1 and M_2 onto a coarser shared substrate — is a case in point, since the merged ambient is inhabited and

	const.	gram.-sel.	pointing	substrate
Falsity	obj R2/R4	cn R1-aug	obj R1-pure	cn R4^{A1}
Incompleteness	obj R1-pure	cn R1-aug	obj R1-pure	cn R1-aug^{A1}
Inaccuracy	obj R2^{A3}	cn R2	obj R2	cn R4
Incomputability	obj R2/R4	cn R2^{A2}	obj R2	cn R4
Incongruence	obj R4^{A3}	cn R4	obj R4	cn R4
Incomparability	obj R4^{A3}	cn R4	obj R4	cn R4
Incommensurability	RES^{A3}	cn R4	cn R4^{A4}	obj R4
Incoherence	DEG	cn R1-aug^{A2}	obj R4	cn R4
Inconsistency	DEG	cn R1-aug^{A2}	obj R3 (meta)	cn R2^{A1}
Infinity	obj R3	cn R1-aug	obj R3	cn R2/R4
Non-linearity	obj R4^{A5}	cn R1-aug	obj R4	cn R4

Table 3: Recorded direction and recorded REI route per criterion and DIA operator. Columns are the four DIA operators: constitutive, grammar-selection, pointing, substrate. "obj" and "cn" abbreviate object-directed and constraint-directed (Definition 38). Superscripts name the adapter of Table 2 the cell presupposes. DEG: undefined at genesis. RES: dissolved by its adapter. "R3 (meta)" is read per Remark 54. The pairings are *witnessed*: each was checked against the definitions of Sections 4.1 to 4.6. The table is not derived from a classification theorem, and no completeness claim is made for it (Remark 55, (Q9)).

so is not Σ_0. Second, the proposition does not license the slogan that REI undoes nothing: routes (R1), (R2) and (R4) change the state substantively, and what fails is only the specific claim that any of them, alone or in composition, returns the undifferentiated substrate.

4.7.5. *The separator/route correspondence.* Table 3 records, for each criterion and each DIA operator, one direction and one REI route, the excluded interval being read off the separator as in Remark 44. We call the entry of a cell its *recorded* pairing: the one this paper puts on file for that combination. The word marks a choice and confers no distinguished status — it does not assert that the pairing is unique, that it is the only well-typed one, or that the route discharges the interval without an intermediate construction (the Inconsistency–pointing cell is precisely a case where it does not, Remark 54). The table is a projection: at most one recorded pairing per cell (a pairing may itself record a tie), chosen and recorded, not derived. Three cells record two routes separated by a slash — Falsity and Incomputability under the constitutive operator, Infinity under the substrate operator — where the programme has used both and neither is the more canonical; the slash marks that tie and nothing further.

Two markers need their grounds stated. Cells marked DEG carry a criterion defined on Γ — Incoherence and Inconsistency both are — applied at the constitutive stage, where by the A5 clause of Table 2 no constraint presentation is available: the criterion has nothing to be evaluated against, so the cell is undefined for want of an argument, not for want of a direction. (This is a distinct ground from Proposition 40(iii), which concerns the availability of the constraint-*directed reading*; conflating the two would collapse the criterion axis into the direction axis.) The cell marked RES is dissolved by its own adapter: A3 supplies at genesis the measure whose absence Incommensurability tests, so no excluded interval survives for a route to act on.

Three readings of Table 3 deserve comment.

Remark 53 (What the recorded routes are). *[WITNESSED.]* Three groups of cells, not three rows, carry most of the table's content. In the *object-directed* Infinity cells the recorded route is (R3), and the Hansel–Newton tuple of Remark 35 is exactly what fills it: a finite certificate discharging an infinitary criterion, which is why that remark and those cells describe one mechanism. In the object-directed Incompleteness cells the recorded route is (R1) in its R1-pure form (Definition 48), since naming the undecided statement is what the criterion asks for; the constraint-directed Incompleteness cells record R1-aug instead, because there the withdrawn requirement has to be replaced. Across the structural and measure-theoretic criteria the recorded route is predominantly (R4), where reintegration consists in building a common ambient — an order extension, a product, a lifting to a space on which the map is linear — rather than in refining the one at hand; the exception is the Incommensurability–constitutive cell, which records no route at all.

Remark 54 (Inconsistency and route R3). *[DEFINITIONAL; a scope correction, not a route.]* The Inconsistency row must not be read as certifying $\neg\bot$ from an inconsistent Γ, which is unavailable. What the cell records is a certificate of the consistency of a *revised* presentation Γ', obtained by an antecedent R1-aug act (Definition 48) that withdraws the offending constraint. The composite is R1-aug followed by a certificate at the meta level; the abbreviation "R3 (meta)" in Table 3 stands for that composite and not for a route applied to Γ itself.

Remark 55 (What the table does not claim). *[Scope statement.]* Table 3 is a catalogue disciplined by the typing of Sections 4.7.1 to 4.7.4, not a classification result. Three questions it leaves open are: whether the criteria of Table 1 are independent, or whether some pair collapses under a suitable reduction; whether the five adapters of Table 2 suffice for every criterion one might add; and whether a completeness statement — every well-typed (criterion, operator, direction, route) quadruple is an instance of a recorded pairing — can be proved. We record these as open (Section 8).

4.7.6. *Deployment modes: non-excisive classification.* The calculus so far types *what* an operation is, *where* it is applied, and *how* its outcome is valued. One datum remains: what the application does to the spectrum itself. Two deployment effects must be distinguished, and neither is a new operator.

Definition 56 (Deployment mode)**.** *[DEFINITIONAL.]* Let a separating criterion Φ (Definition 43) be applied over a spectrum $\Sigma = (X, \mathsf{Adm}, \mathsf{Exc})$, inducing cells on the admitted material. The application is

- *cutting* when at least one induced cell is typed against the spectrum — moved to, or named in, Exc, or given a forbidden reading. The extent $\bigcup \mathsf{Adm}$ shrinks; boundary, provenance, and possible reintegration obligations arise as in the preceding subsections.
- *classifying* when every induced cell remains admissible: the three extents $\bigcup \mathsf{Adm}$, $\bigcup \mathsf{Exc}$, and the residue are unchanged, and the admissible *family* is refined to the induced cell family. This is the trivial case of the constitutive act (Section 4.1, the case $\pi^{-1}(0) = \varnothing$), iterated: no new primitive is involved, and the induced family is exactly the Adm_1-family of that definition.

The mode is an input of the act, not a function of the criterion: the same Φ supports both deployments (Remark 58). Direction and valence (Definitions 38 and 39) do not type this datum: a classifying act leaves (X, Γ) fixed, so its valence comparison is undefined rather

than neutral, and a description-directed target is a candidate extension of the direction axis, not asserted here.

Proposition 57 (Non-excisive distinction)**.** [WITNESSED, by exhaustive enumeration in the finite validation model; the general form is a schema, not a theorem.] *In the finite validation model of this subsection's claims, an adopted classification strictly refines the induced description if and only if the criterion's trace on the admitted extent is not already expressible as a union of existing cells; in every case the three extents are unchanged. Distinction is produced and nothing is excised: the excluded component receives no member, and no excluded interval arises from the act.*

Remark 58 (Same criterion, both deployments)*. [DEFINITIONAL in the model; the content is the freedom of the mode input.]* For every criterion nontrivial on the admitted material, the classifying deployment leaves Exc unchanged while the cutting deployment of the *same* criterion changes it. A separating criterion is therefore not intrinsically destructive; whether its use is classificatory or excisive is contextual. DIA is an algebra of distinction, of which exclusion is one deployment.

Remark 59 (Classification and the excluded interval)*. [Scope discipline.]* The complement of a class is not an excluded interval. "Not selected", "in another cell", "excluded", and "verifier-undecided" are four different types, and only the last is the object the pointing convention of Remark 44 reads as IE: under a total criterion the undecided block $B^{?}_{\Phi}$ is empty, so the convention is vacuous there, and its legitimate object is the genuinely partial verifier; the one-sided case is already handled at the pointing level by Definition 6 (the parenthetical two-sided reading, under which the one-sided IE is empty). A classifying deployment can thus produce a nontrivial refinement with IE $= \varnothing$ throughout.

Remark 60 (Order behaviour of the two modes)*. [WITNESSED (order-freedom, exhaustively; the refuting instance) in the finite validation model.]* For *state-independent* criteria — fixed extensions, applied passively — the induced description depends only on the *set* of criteria applied: sequential classification computes the common refinement, a semilattice meet, so order and interleaving are immaterial. For *state-dependent* rules (a later classifier computed from the current cells) the commutation question is ill-posed at the level of criteria — the two orders deploy different criteria — and the well-posed operator-level statement fails on a finite witness. Cutting is intrinsically state-dependent (the second cut's excluded set depends on the first); in the validation model its final extents nevertheless commute under complement-exclusion, and the order sensitivity is carried entirely by the provenance record — consistent with the path-dependence results of the cut algebra.

Remark 61 (Interface to symbolic reification)*. [Architecture statement; each arrow carries the status noted.]* Classification is upstream of symbol formation, across an interface with strict level separation:

$$\Sigma \xrightarrow{\text{classification}} \text{cells} \xrightarrow{\text{mining}} R^{\star} \xrightarrow{\text{reification (MCK)}} \text{token } s^{\star},\ \text{grammar } G'.$$

Mining here means the inspection of the refined description for stable regions not yet named — a candidate region $R^{\star}$ is not a concept, and no semantic-validity criterion is supplied by the classification layer. Reification and token genesis belong to the companion MCK embedding framework [Bra+26], whose verifier transition and controlled dimensional augmentation gate the step on two inputs left open there (the verifier's acceptance and the availability of a candidate operator); classification is one *sufficient* producer of gate-satisfying configurations

(a mined region, token-less and inexpressible in the current grammar), and is not necessary for it. One feedback arrow is witnessed: a minted token, adopted as a predicate, makes previously inexpressible distinctions expressible. No claim is made that iterating the interface reaches every position or terminates; the companion's fixpoint-vs-frontier question remains open. The expressibility layer used on the classification side — cells as indiscernibility classes, expressible regions as unions of cells — is rough-set definability [Paw82], imported as a known tool; conceptual-space geometry [Gär00] offers a neighbouring structural criterion for the validation slot deliberately left open, and representation-learning practice reaches neighbouring territory with the levels collapsed [HVZ19] or with the concept layer supplied by supervision [Koh+20]. The contribution of this subsection is not classification, refinement, or clustering individually; it is their typed placement inside the DIA distinction calculus and the explicit, level-separated interface to symbolic genesis. Figure 4 draws the two deployments and the interface.

4.8. **A theorem-target: DIA-freedom and topological clopenness.** The following statement is a target for structural theory, not a proved result at the time of writing. It recovers, under ISG, the classical clopen ⇔ monitorability characterisation [Bra26e; BLS07] and generalises it to Atomic-Geometry spectra beyond the Σ^ω-topological case.

Conjecture 62 (D1: DIA-freedom via clopenness)**.** *Let Σ be an Atomic-Geometry spectrum equipped with a topology in which* Adm *and* Exc *are Borel-measurable. For any statement s in the domain of a grammar G,*

$$\begin{aligned} \mathrm{DIA}_{\mathrm{point}}(s; G, \Sigma) = \emptyset \quad &\Longleftrightarrow \quad \rho_G(s) \textit{ is } \text{clopen } \textit{in } \Sigma \\ &\Longleftrightarrow \quad s \textit{ is monitorable in the strong sense at } (G, \Sigma) \\ &\Longleftrightarrow \quad s \textit{ is sayable on any finite prefix.} \end{aligned}$$

Equivalently: a partition of Σ that admits $\mathrm{SP}_G(s)$ and excludes $\mathrm{SR}_G(s)$ has an empty excluded interval iff the carried property is clopen; every non-clopen property carries an irreducible IE*;* REI *can empty* IE *only by supplying a matching finite-confirmation certificate ((R3)).*

Remark 63 (Witnessed instance). The horizon note of [Bra26e] establishes, inside the ltl_classes canon, that the L/K residue *is* the formal content of the excluded interval for temporal properties over Σ^ω. This is the one witnessed instance of Conjecture 62 at the time of writing; D1 as universally quantified over Atomic-Geometry substrates is open.

Remark 64 (Relation to the classical monitorability characterisation). On the specific substrate Σ^ω with the standard topology, the equivalence $\rho_G(s)$ clopen ⇔ s monitorable recovers the impartial-anticipation characterisation of [BLS07]: a property is monitorable in the classical sense exactly when its violation and its satisfaction can each be detected by a finite prefix, which topologically forces the property to be clopen. Conjecture 62 does not strengthen this classical statement on Σ^ω; its ISG-native content, beyond re-expression, consists of *(a)* the generalisation to Atomic-Geometry substrates without an assumed infinite-trace topology (the conjecture asserts that clopenness in the substrate's own topology — whenever such a topology is definable from the atomic generation — characterises DIA-freedom in the same way), and *(b)* the reflexive reading (Section 6), where DIA-freedom of a positioning statement on the admissible-readings spectrum $\Sigma(C_P)$ is characterised by the same clopenness criterion, opening the monitorability lens to statements about scientific

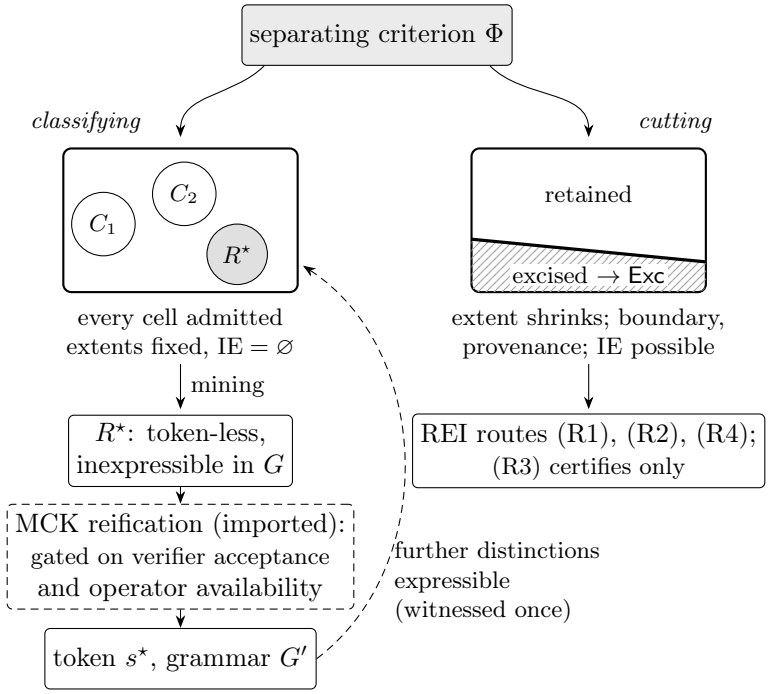


Figure 4: The two deployment modes of one separating criterion (Definition 56). Left: the classifying deployment refines the admitted material of an intact spectrum into cells, all retained — classification excludes no semantic material and produces no excluded interval; mining then isolates a stable, still-unnamed region $R^\star$. The dashed block is owned by the companion MCK embedding framework [Bra+26]: reification and token genesis are gated there on two inputs left open (verifier acceptance, operator availability), and symbolic genesis is a downstream MCK process, not an effect of classification. Right: the cutting deployment of the same criterion changes the excluded component, with the boundary, provenance, and reintegration discipline of the preceding subsections. The dashed return arrow records the one witnessed feedback: a minted token can make further distinctions expressible; no loop theorem is drawn, and nothing here restricts criteria to linear separators.

contributions that classical monitorability does not address. Both (a) and (b) are open beyond the one witnessed instance.

4.9. **The aftermath of the constitutive cut: the (M_1, C, M_2) separation.** The generative cycle closes with a synthesis that is the paper's payoff and its bridge to the MCK line. A constitutive DIA event (Definition 15) does not merely name a gap in a substrate given elsewhere; it *leaves a structured aftermath.* The partition act $\pi : X \to \{+, -, 0\}$ produces, in a single stroke, two stabilised admitted domains and the excluded interval between them. Naming these three residues is the last constitutive datum of the cycle, and it is where ISG meets the MCK region calculus.

Definition 65 (The (M_1, C, M_2) separation)**.** *[DEFINITIONAL; an identity of the constitutive datum, not a new construction.]* Let (π, G_1, Σ_1) be a constitutive DIA event at an undifferentiated substrate Σ_0 (Definition 15), with distinguished labelling statement s_+ and partition-dual s_-. The *aftermath* of the cut is the ordered triple

$$(M_1,\ C,\ M_2)\ :=\ \big(\mathrm{SP}_{G_1}(s_+),\ \mathrm{IE}_{G_1}(s_+),\ \mathrm{SR}_{G_1}(s_+)\big)\ =\ \big(\pi^{-1}(+),\ \pi^{-1}(0),\ \pi^{-1}(-)\big),$$

where $M_1 = \mathrm{SP}_{G_1}(s_+) = \pi^{-1}(+)$ is the *admitted-+* stabilised domain (the projective solid the statement lands in), $M_2 := \mathrm{SR}_{G_1}(s_+) = \rho_{G_1}(s_-) = \pi^{-1}(-)$ is the *admitted-−* residual domain (s_- read as $\neg s_+$ in the sense of Definition 6), and $C = \mathrm{IE}_{G_1}(s_+) = \pi^{-1}(0)$ is the excluded interval between them. In the constitutive act both sides are admitted ($\mathsf{Exc}_1 = \varnothing$): M_2 is the *residual* admitted side in the two-solid sense of Remark 13 — SR as the autonomous alternative rather than as an Exc-region. Consequently

$$(M_1, C, M_2) = (\mathrm{SP}, \mathrm{IE}, \mathrm{SR})$$

is a definitional identity, with C placed *between* the two stabilised sides: the aftermath *is* the SP–SR–IE canon (Section 3) read off the constitutive act, not a further theorem about it.

Remark 66 (C is the high-entropy space-of-difference)*. [WITNESSED convergence with MCK's opacity predicate; grounded on [BAI26] and Lemma 21.]* Of the three residues, $C = \mathrm{IE}_{G_1}(s_+)$ is the distinctive one: it is the region the partition leaves neither on the admitted-+ nor on the admitted-− side — the *symbol not yet resolved*, the maximal-denotation-multiplicity middle whose points sustain both readings and settle into neither. In the entropy reading of the SP–SR–IE canon, M_1 and M_2 are the ordered, low-entropy stabilised sides and C is the high-entropy interval-between: the un-understood space-of-difference of the cut.

This C *converges* with the opacity predicate C of the MCK framework (Definition 2 of [BAI26]): MCK's C is exactly the artifact "epistemically present but not yet semantically integrated", which on MCK's semantic/entropy axis is the "high-entropy" ("opaque, unstructured") side rather than the ordered low-entropy side, i.e. "symbolic reference not yet resolved into structured meaning". The convergence is not first asserted here: the genealogy of Remark 20 already identifies MCK's C as the product of the constitutive stage, with the canonical region-to-predicate bridge PROVEN in Lemma 21 and the MCK coincidence WITNESSED under the constitutive-adequacy hypothesis. The present subsection records the consequence for the aftermath: the cut's middle residue *is* its high-entropy, un-understood zone, and that zone is MCK's C.

Remark 67 (Naming: (M_1, C, M_2) versus MCK's $(\mathsf{M}, \mathsf{C}, \mathsf{K})$)*. [Naming discipline (G7); the outer triple correspondence is CONJECTURED, only C is WITNESSED.]* The shared letters invite a false identification that we block explicitly.

- M_1 **and** M_2 **are the two sides of one cut**, i.e. the two stabilised admitted domains SP and SR of a single constitutive act. They are *not* two copies of, nor instances of, MCK's single predicate M (*Meaningful*, Definition 1 of [BAI26]). The subscripts index sides of a partition, not epistemic predicates.
- The triple (M_1, C, M_2) and MCK's triple $(\mathsf{M}, \mathsf{C}, \mathsf{K})$ **share letters but are different objects**. Only the middle term genuinely corresponds: cut-C = IE = MCK-C (Remark 66, WITNESSED). The outer correspondence $M_1/M_2 \leftrightarrow \mathsf{M}/\mathsf{K}$ is a BRIDGE / ANALOGY, **not an identity**, and is tagged CONJECTURED: the ISG–DIA–MCK bridge still has open

gates — the role of the functional predicate K (G5) and the naming (G7). In particular $M_2 = \mathrm{SR}$ is a *stabilised side of the cut*, whereas MCK's K is the orthogonal *functionality* axis (Definition 3 of [BAI26]); the two are not the same object and no identification is asserted.
- The numerical echo — between MCK's seven-region cube and any seven-fold refinement of the two sides of the cut — is **coincidental**. We assert no structural $7 = 7$ correspondence; a coincidence of counts is not evidence of a shared partition.

Remark 68 (The cut as semantic-body separation). *[HEURISTIC; interpretive bridge to the semantic-body instance, no new formal claim.]* The (M_1, C, M_2) separation is what a constitutive cut performs on a *semantic body* (Section 5.5). There the two agents' product substrate is separated by a tuning-driven cut into two stabilised sides — the regions on which each agent's reading is admitted — with C the high-entropy difference-space on which the two readings have not yet been made co-referential. A substrate-REI via canonisation ((R4)) is then precisely the suture that collapses C, merging M_1 and M_2 onto a shared coarser substrate. The aftermath thus names, in one triple, both what the cut leaves standing and what the suture must repair.

Remark 69 (Polarity of the cut is contextual). *[DEFINITIONAL, discharged in Section 4.7; polarity is a function of the target's constraint-state, not a fixed sign.]* Whether a constitutive cut is a loss or a gain is not decided by the cut operation alone; it depends on the constraint-state of the target. On a *coherent whole*, a cut fragments an integral substrate and its polarity is negative. On an *over-constrained* target — one whose admissible region has collapsed so far that agent freedom is eliminated — a cut that removes the excess constraint re-opens the admissible region, and its polarity is positive (a liberating, freedom-restoring cut). Polarity is therefore evaluated against a coherence/freedom baseline, not asserted as a fixed property of DIA. The typed-criterion account of the cut and of its polarity is given in Section 4.7: the two cases distinguished here are the object-directed and the *subtractive* constraint-directed readings of Definition 38 — subtractive specifically, since an additive constraint-directed act shrinks the admissible region rather than re-opening it — the "re-opening" of the second case is Remark 37, and the baseline-relativity of the polarity itself is Definition 39 together with Proposition 40(i).

5. Programme Substrates as ISG Instances

We collect substrates already studied independently across the programme and exhibit each as an ISG realisation. The purpose is not to re-prove any substrate's results; it is to make the ISG discipline visible on cases the reader will already recognise. Where a companion manuscript is under preparation the citation is marked accordingly.

5.1. **ADKL hyperspheres (Solid Integral).** *Mode.* Solid Integral. *Spectrum.* $X = \mathbb{R}^n$ with Euclidean topology and Lebesgue measure. *Atoms of admissibility:* finite families of hyperspheres $\{(c_i, r_i)\}$; admissible regions are Boolean combinations of interior / exterior / boundary-zone parts of the primitive balls; excluded regions are the same for their negations. *Grammar.* A predicate p maps to the interior of its associated ball; $\neg p$ to the exterior; the epsilon-thickened boundary ∂_ϵ is either shared by both (the Kleene Unknown zone) or unaddressed (a genuine pointing DIA). *Excluded interval.* $\partial_\epsilon A_i$ of every primitive; the

width of the excluded interval is measurable, and near-tangency between two predicates (width ≈ 0) is a fragility indicator [Bra26a].

5.2. **The MCK region cube (Atomic Geometry).** *Mode.* Atomic Geometry (with a mixed reading: the seven regions also carry a solid-integral interpretation as measurable partition cells when the M/C/K predicates are lifted to a measure space). *Spectrum.* $X = \{0,1\}^3$ (the M/C/K cube), plus its seven canonical regions $R_1, \ldots, R_7$ and the boundary zones between them. *Atoms.* The three MCK predicates M (Meaningful), C (Contained), K (functional / generative). *Grammar.* An artifact x is classified into a triple $(m_x, c_x, k_x) \in \{0,1\}^3$, from which its canonical region and any relevant boundary neighbourhood are computed. *Excluded interval.* Boundary zones between adjacent regions — the transition surfaces on which promotion from one region to another may take place without explicit epistemic justification — are exactly the pointing DIA of this substrate [BAI26].

5.3. **The S/C/L/K four-family (Atomic Geometry).** *Mode.* Atomic Geometry. *Spectrum.* The 16-cell membership partition of Σ^ω-properties under the four topological modalities Safety, Cosafety, Liveness, Coliveness. *Atoms.* The four modalities $\mathsf{S}, \mathsf{C}, \mathsf{L}, \mathsf{K}$ themselves, each a Borel subclass of $\mathcal{P}(\Sigma^\omega)$; the modalities are the atoms, not the 16 cells. *Generation rules.* Finite intersection and finite union of the four modalities, plus the atomic Boolean operations, generate the 16-cell membership partition as the lattice of joint memberships. *Admissibility structure.* Adm is the family of Boolean combinations of atoms whose intersection is non-empty; Exc is the family whose intersection is empty. *Grammar.* An LTL formula φ is placed into its canonical cell (clopen if $\mathsf{S} \cap \mathsf{C}$; empty-interior if $\mathsf{L} \cap \mathsf{K}$; etc.) by reading the operator skeleton. *Excluded interval.* The strong non-monitorable cell ($\mathsf{L} \cap \mathsf{K}$) is the substrate-level excluded interval of any decomposition that attempts finite verdicts on infinite-trace properties; the horizon note of [Bra26e] formalises this identification. The one witnessed instance of Conjecture 62 lives here.

5.4. **Touch and Crossing (dual Atomic Geometry).** *Mode.* Atomic Geometry. *Spectrum.* The capability lattice $\kappa = (w, b, k)$ over window, bits, lookahead, joined via the augmented-automaton join to $\kappa = (3, 2, 1)$. *Atoms.* The two primitives: crossing $\kappa_{\mathrm{X}} = (1, 1, 0)$ and touch $\kappa_{\mathrm{T}} = (3, 0, 1)$; their join $(3, 1, 1)$; the general augmented $(3, 2, 1)$. *Grammar.* A threshold-monitor specification is placed at its minimum capability triple. *Excluded interval.* A specification presented in one primitive that would require a capability beyond the primitive's κ — e.g. a crossing-only spec that in fact requires lookahead — is a substrate DIA between the crossing and touch substrates and can only be reintegrated by canonisation into the $(3, 2, 1)$ join [Bra26g].

5.5. **Semantic body: two-agent, three-channel substrate.** *Mode.* Mixed — Atomic Geometry on the epistemic-side factor (MCK cube inherited); Solid Integral on the observation-side factor (metric distances between channel readings).

Formal setup. A semantic body is a two-agent structure in which each agent $A \in \{a_1, a_2\}$ maintains three channels: SELF (internal epistemic state), COMMUNICATION (dyadic exchange between a_1 and a_2), and BALL (an external shared referent against which both agents interpret their observations). The spectrum is a product $\Sigma_{\mathrm{sb}} = \Sigma_{\mathrm{self}} \times \Sigma_{\mathrm{comm}} \times \Sigma_{\mathrm{ball}}$ with admissibility structure inherited channel-wise: Σ_{self} inherits from the MCK cube of

each agent (Section 5.2); Σ_{comm} inherits from a two-agent alignment sublattice reading of the intersection $\text{MCK}_{a_1} \times \text{MCK}_{a_2}$; Σ_{ball} inherits a Solid-Integral observation-metric on the shared referent, from which primitive regions (thresholded distance neighbourhoods) are drawn.

Grammar. A tuning depth δ selects an observation window that reads each channel with a fixed granularity; the grammar G_δ is parametric in δ. Different tuning depths determine different candidate references for the same behaviour, so the semantic body is intrinsically a *substrate*-DIA setting: varying δ varies Σ_{sb}'s effective admissibility, and reference-preserving translations (Definition 28) between G_{δ_1} and G_{δ_2} need not exist for every pair.

Excluded interval. A tuning move on Channel 3 (BALL) can, under sufficient parameter change, force the Channels 1–2 product $\Sigma_{\text{self}} \times \Sigma_{\text{comm}}$ to collapse onto the region where the two agents' MCK states agree — a *substrate* collapse (a merging of previously-distinguished regions of the substrate itself), not a pointing collapse (a narrowing of a reference inside a fixed substrate). The conjectured channel-collapse theorem of [Bra26f] is, in ISG terms, a specific substrate-REI via canonisation ((R4)): the tuning move canonises the two-agent grammar into a shared coarser substrate on which the two agents' MCK states become co-referential.

Load-bearing role. The semantic body is the primary programme instance in which substrate indeterminacy dominates over pointing indeterminacy. ISG's substrate-mode operator (Definition 29) is load-bearing for the semantic body's analysis, and downstream work on channel-collapse, tuning-driven state changes, and multi-agent alignment can be posed and reasoned about directly in the framework introduced here.

5.6. **Epistemic body: shared witness register under substrate DIA.**

Remark 70 (Witness body vs epistemic body). *[DEFINITIONAL distinction between single-agent witness body and multi-agent epistemic body.]* Earlier programme notes distinguish two related-but-distinct substrates that we keep separate here: (i) a *witness body* WB_a attached to a single agent a, whose reference values are KNOWN_a-tagged and whose admissibility is a promotion ladder across MCK regions R_1–R_7 (cf. [BAI26]); (ii) an *epistemic body* $\mathsf{EB}_{(a_1,\ldots,a_n)}$ attached to a multi-agent group, whose reference values live in the shared register that arises from the family of per-agent grammars. The witness body is a single-agent substrate under one grammar; the epistemic body is the multi-agent shared substrate under many grammars. Both are ISG instances in the Atomic-Geometry mode; the distinctive feature of the epistemic body is that substrate DIA is not a corner case but the operator's primary domain of application. This subsection treats the epistemic body; the witness-body case reduces to Section 5.2 plus a KNOWN_a-labelling on the region cube and does not introduce distinct ISG machinery.

Mode. Atomic Geometry with substrate indeterminacy as the distinguishing feature (each agent's per-agent admissibility predicate is itself a candidate substrate, and the shared substrate is not canonically given).

Formal setup. An epistemic body is a multi-agent structure $(A_1, \ldots, A_n)$ in which each agent A_i contributes its own grammar G_i over a shared statement space $\mathcal{S}$ and its own reference $\rho_{G_i}(s) \subseteq X_i$ into its own spectrum $\Sigma_i = (X_i, \mathsf{Adm}_i, \mathsf{Exc}_i)$. The *shared substrate* $\Sigma_{\text{shared}} = (X_{\text{shared}}, \mathsf{Adm}_{\text{shared}}, \mathsf{Exc}_{\text{shared}})$ is not canonically determined by the individual Σ_i: it is the object of collective determination via the reference-preserving translations of Definition 28, and its structure depends on which per-agent grammars agree.

Grammar. A meta-grammar G_{shared} over Σ_{shared} is definable when the family $\{(G_i, \Sigma_i)\}_{i \le n}$ is substrate-coherent at every relevant s in the sense of Definition 29; otherwise the shared substrate is only partially determined, and $\text{DIA}_{\text{sub}}(s; \{(G_i, \Sigma_i)\})$ is the class of ordered pairs on which no reference-preserving translation exists — exactly the obstruction that prevents a canonical shared grammar from arising.

Excluded interval. Exactly the substrate DIA of Definition 29. The epistemic body is the paradigmatic case in which substrate DIA is not a corner case but the operator's primary domain of application: it *is* the mechanism by which multi-agent disagreement over the shared substrate becomes a formal object.

Adversarial reading. When one agent's grammar-boundary is discoverable by another via observed rejection/acceptance patterns — the "epistemic body as counterexample-oracle" phenomenon — the substrate DIA becomes an oracle: the observer can approximate the private grammar of the observed by iterating queries near its boundary. This is an adversarial ISG-REI via canonisation ((R4)), run by an observer against a target rather than by an author against their own grammar. It is the adversarial dual of the benign grammar-extension REI ((R1)); both operate on the substrate DIA but with opposite intent. Full formalisation of the adversarial protocol, including the associated defensive mechanisms (transcript binding, statement binding, semantic binding, bridge resistance), is deferred to future work (Section 8); the observation here is that ISG names the operation, its operand, and its adversarial dual within a single framework.

Remark 71 (Semantic body and epistemic body as ISG duals). The semantic body (Section 5.5) and the epistemic body (this subsection) instantiate ISG from complementary sides. Semantic body: a small-agent (two-agent) product substrate with a *grammar parametric in a tuning depth* — substrate DIA arises from varying δ across a single family of candidate grammars. Epistemic body: a multi-agent shared substrate with *multiple distinct grammars* $\{G_i\}$ one per agent — substrate DIA arises from disagreement between the G_i themselves. Their joint use in downstream programme work supports a *triangulation methodology*: for a target concept c under introduction, the semantic body's geometric admissibility (tuning-parametric substrate) and the epistemic body's shared-witness admissibility (multi-agent substrate) jointly discipline the introduction of c into the programme's reference space. Substrate coherence between the two sides is a precondition for the concept to be programme-referenceable. A full development of the semantic-body / epistemic-body triangulation is left as an ISG application for future work.

5.7. **Two-register dynamics: ISG in motion.** *Mode.* Mixed — Atomic Geometry on Σ_{state}; Solid Integral (measure-carrying) on Σ_{accum}. *Spectrum.* A product $\Sigma_{\text{state}} \times \Sigma_{\text{accum}}$ of a spectrum whose points are (semantic) state configurations and a spectrum whose points are (epistemic) accumulator values. *Grammar.* A rule split into a region-move on Σ_{state} (an instantaneous change of admissibility) coupled to a monotone accumulation on Σ_{accum}; a firing rule triggers when the accumulator crosses a Hansel–Newton threshold. *Excluded interval.* The gap between what the state-move geometrically implies and what the accumulator has so far witnessed is the pointing DIA of this dynamic system; the two-register integrate-and-fire mechanism is a REI-in-motion. The full dissociation of the two routes to semantic state-change (autonomy vs. contact-absorption) is left open here (Section 8, Q6).

6. Reflexive ISG: Statements about a Scientific Contribution

We now instantiate the framework where its structural payoff is sharpest for the scientific writer: statements a paper makes about its own contribution.

6.1. **Setup.** Fix a paper P and let C_P denote the scientific contribution P represents. Consider the space of *admissible readings* of C_P: the interpretations under which C_P can be legitimately understood by later readers. This space is not an unstructured bag: it is a spectrum $\Sigma(C_P)$ whose regions are the reading-classes a grammar for scientific-contribution-identity would recognise. The catch is that the grammar for scientific-contribution-identity itself is under-determined: different disciplinary communities, different venues, and different eras use different G to name the regions of $\Sigma(C_P)$.

6.2. **Scope: truth-apt positioning.** The reflexive analysis of this section is restricted to positioning statements that are *truth-apt* in the sense of classical semantics: statements of the form "C_P instantiates class K", "C_P does not belong to F", "C_P is best read under frame F" are candidates for reference into an admissible-readings spectrum $\Sigma(C_P)$, because they schematically admit a right/wrong verdict once the grammar for scientific-contribution-identity is fixed. Statements that are pure speech acts in the sense of [Aus62] and [Sea69] — explicit performatives ("we hereby declare X"), endorsements ("this is the right approach"), promises, requests — fall outside the present scope: they do not *purport* to name a reference-region, and treating them as ISG operations would import truth-conditional content they do not carry. A separate analysis of speech-act positioning is left as future work.

6.3. **Positioning statements as ISG operations.**

Proposition 72 (Positioning statements are ISG operations)**.** *Let s be a positioning statement of the form "C_P is K", "C_P is not K", "C_P belongs to class K", or "C_P must be read under frame F". Under any grammar G for scientific-contribution-identity, s is an* ISG *operation on $\Sigma(C_P)$: it names a reference $\rho_G(s) \subseteq \Sigma(C_P)$ (the admitted reading-class), an excluded region $\mathrm{SR}_G(s)$ (the forbidden reading-class, if s is negative), and leaves behind an excluded interval $\mathrm{IE}_G(s)$. The composition of several such statements is a sequence of* ISG *operations $D_1; D_2; \ldots; D_n$ on $\Sigma(C_P)$, not a single one.*

Proof sketch. Fix any grammar G for scientific-contribution-identity that is defined on s — for example, the grammar induced by a target venue's typology of contributions, or the grammar induced by a fixed disciplinary taxonomy. Under G, the statement s receives a value $G(s) \in \mathsf{Adm} \cup \mathsf{Exc}$ by Definition 4, which by definition is $\rho_G(s)$; this is the admitted (or explicitly excluded) reading-class. The residual $\mathrm{IE}_G(s)$ is then computed by Definition 27. All three quantities exist for any such G; the proposition asserts existence, not uniqueness, so the schematic derivation suffices. Composition ($D_1; \ldots; D_n$ for a sequence of positioning statements) follows because G's domain is a set, and a sequence of statements produces a sequence of references, each computed independently under the same G. The choice of a canonical grammar for scientific-contribution-identity is a substrate question (see Section 8), not a proof-content question. □

Remark 73 (Double indeterminacy). The reflexive case is the paradigmatic setting of double indeterminacy. Pointing indeterminacy is present: for a fixed G and $\Sigma(C_P)$, the residual $\mathrm{IE}_G(s)$ is the reading-slack that the positioning statement leaves unaddressed. *But substrate indeterminacy is also present*, and is often dominant: no canonical grammar for scientific-contribution-identity is agreed upon. A disciplinary grammar, a venue grammar, a research-programme grammar, and a historical-tradition grammar all produce different $\Sigma(C_P)$, each with a different admissibility structure. The substrate DIA across these candidate grammars is the reason "what a paper is" disputes are structurally under-determined, not merely a matter of preference.

6.4. **Independence from evaluation.**

Proposition 74 (Independence)**.** *The classification of a positioning statement s as an* ISG *operation on $\Sigma(C_P)$ (Proposition 72) is independent of whether s is correct, useful, necessary, or appropriate for a particular venue. Symbolically:*

$$\mathsf{ISG}\text{-}op(s; G, \Sigma(C_P)) \not\Rightarrow \mathrm{Correct}(s),\ \mathrm{Useful}(s),\ \mathrm{Necessary}(s),\ \mathrm{VenueAppropriate}(s, V).$$

Correctness, usefulness, necessity, and venue-appropriateness are subsequent REI-*evaluations of the classified operation, not conditions for the classification itself.*

Proof. Structurally: the classification asserted by Proposition 72 concerns only whether s receives a reference $\rho_G(s)$ under G in $\Sigma(C_P)$. The four evaluations (correctness, usefulness, necessity, venue-appropriateness) each require a further object beyond $\rho_G(s)$: a target of correctness (an independent reference for the "true" reading of C_P); a use-context; a notion of unavoidability; a venue's own grammar and admissibility norms. Each evaluation is a relation between $\rho_G(s)$ and one of these further objects, and can be applied only after $\rho_G(s)$ has been identified. The classification is therefore prior to each evaluation and does not entail any of them.

A non-obligatory schematic reading suggests that each evaluation corresponds to one REI route: correctness pushes toward substrate refinement ((R2)); usefulness toward grammar extension for a target application ((R1)); necessity toward finite-confirmation of unavoidability ((R3)); venue-appropriateness toward canonisation against the venue's own grammar ((R4)). We present this correspondence as an interpretive heuristic, not as part of the proof; the independence claim does not depend on it. □

Remark 75 (Confusion of levels). An author or reviewer who rejects a positioning statement on the ground of correctness, usefulness, or necessity is running REI on the underlying DIA, not falsifying the classification itself. When the levels are confused — "that positioning is wrong, therefore this is not really ISG" — the response is to name the levels explicitly and continue.

6.5. **Payoff for programme papers.**

Corollary 76 (Core/positioning separation)**.** *Any scientific paper P's contribution C_P admits a decomposition into*

(i) *its* invariant scientific core — *formal objects, definitions, assumptions, proofs, experiments, and falsifiable scope conditions — to which the paper's declared formal-scope invariants apply and must remain sharp; and*

(ii) *its* optional interpretive positioning — *the disciplinary label, the excluded neighbouring frame, the venue voice — to which Proposition 72 and Proposition 74 apply, and which is trimmable, tunable, or omissible without weakening (i).*

The core is the target of the formal machinery of the paper. Positioning is a separate object, subject to the ISG *discipline described above.*

Corollary 76 is the citable payoff of Section 6. Subsequent work may reference it to justify (a) tightening or removing extensive positioning digressions when they do not advance the invariant core, without weakening the paper's technical claims; (b) treating a venue's positioning demands as REI requests over the paper's default DIA; (c) naming as REI rather than refutation the response to a reviewer or reader who contests the paper's positioning.

7. Related Work

ISG sits in a neighbourhood of frameworks in which meaning is handled through geometric, category-theoretic, or intensional substrates rather than through Boolean propositions alone.

Classical intensional semantics. The statement / spectrum / reference triple has evident predecessors in Frege's sign / sense / reference distinction [Fre92] and Carnap's intension / extension partition [Car47]. Both classical accounts take the mediating function (Fregean sense; Carnapian intension) as a fixed determinant of a canonical target (reference; extension); neither treats the target substrate as a variable of the analysis. ISG's minimal addition on top of the classical picture is to promote both the grammar G (playing the role of sense/intension) and the spectrum Σ (playing the role of the target of reference/extension) to objects of research, and to name substrate indeterminacy (Definition 29) as the mode of ambiguity neither Frege nor Carnap models without stepping outside their formal apparatus. Pointing indeterminacy (Definition 6) recovers a formal parallel to Frege's own concern with sense-vs-referent gaps for empty singular terms.

Geometric logic and locales. Vickers's programme of *topology via logic* [Vic89] and Johnstone's *Elephant* [Joh02] develop the tradition in which a proposition is a point in a locale and the semantics is intrinsically topological. ISG's spectrum, when equipped with the topological structure of Definition 2, is a locale-like object; the admissibility structure Adm and the excluded structure Exc provide an extra bookkeeping layer that a bare frame does not, and that supports the SP/SR/IE trichotomy (Section 3). ISG's distinctive move relative to locale semantics is the substrate indeterminacy operator, which naturally lives across candidate locales rather than inside a single one.

Sheaf semantics and topos theory. Sheaf-theoretic semantics organises the contexts under which a formula is locally satisfied via a Grothendieck site; the canonical reference is [MM94]. Awodey's later work on sheaf representations for topological semantics [Awo00] sharpens the correspondence between sheaf-theoretic satisfaction and topological interpretation. A grammar in ISG plays a role analogous to reindexing along a site morphism when the underlying spectra carry sheaf-theoretic structure; the pointing DIA over a fixed sheaf corresponds to the local failure of a formula to be forced at a stage, and substrate DIA corresponds to the obstruction to a common refinement of two sites (Definition 29). Where sheaf semantics presupposes a Grothendieck topology, ISG does not: the framework

accepts spectra with weaker structure (metric, graph, plain topological) at the cost of losing the classifying-topos machinery. Genuine additions of ISG over sheaf semantics are (a) the explicit SP–SR–IE trichotomy inside a single admissibility structure, (b) the reflexive-application methodology of Section 6. Where the substrates ARE topos-theoretic, sheaf semantics is the more powerful tool and ISG is a book-keeping layer on top.

Girard's geometry of interaction. Girard's *geometry of interaction* [Gir89] interprets proofs as operators on an ambient geometric space. Where Girard's spectrum is fixed (a Hilbert space of tokens), ISG's spectrum is the object of research: substrate indeterminacy is a first-class datum, not a residual of the formalism.

Nominal sets and hyperdoctrines. Gabbay and Pitts's nominal sets [GP02] formalise a grammar as an equivariant action; Lawvere's hyperdoctrines cast interpretation as reindexing between fibred categories [Law69]. ISG borrows the insight that grammar is structural rather than accidental, but does not require the reindexing to be a functor.

Monitorability and finite-confirmation. The classical monitorability boundary of Bauer–Leucker–Schallhart [BLS07] is, under ISG, an instance of the clopen theorem-target (Conjecture 62) on the Σ^ω substrate. Bounded model checking [Bie+99], k-induction [SSS00], IC3 / PDR [Bra11; EMB11], and CEGAR [Cla+03] are all instances of REI routes (R1)–(R3). Runtime verification [HR04] is a runtime instantiation of the same routes on the observation-window substrate.

What ISG adds. Positioned honestly against the neighbours above, ISG's contributions are:

(a) The *substrate DIA operator* (Definition 29) — naming as a first-class object the obstruction to translation-preserving comparison across candidate (grammar, spectrum) pairs. Sheaf semantics handles cross-site issues through cover refinement, but does not name the failure of common refinement as a distinguished operator; classical intensional semantics does not treat substrate variation at all. The substrate DIA is a genuine primitive of the framework.

(b) The *reflexive* ISG *application* (Section 6) — exposing a paper's own positioning statements as ISG operations on the contribution's admissible-readings spectrum, subject to the same substrate/pointing indeterminacy bipartition as any other ISG statement. This methodological payoff is not present in any of the neighbours above.

(c) The *systematisation of DIA/REI as derived discipline* (Section 4) — collecting under a single formalism the excluded-interval calculus that classical abstraction-refinement, monitorability-boundary, and finite-confirmation literatures develop piecewise. This is a unification, not new mathematical content beyond the pieces.

(d) The Solid $\leftrightarrow$ Atomic distinction (Definitions 9 and 10) as a research direction — suggesting a duality theorem (Remark 12) that would connect the extensional and intensional presentation modes for a well-behaved sub-class of spectra. This is a target for further work, not a result of this paper.

Items (a) and (b) are the paper's substantive contributions; (c) is a systematisation with pedagogical and reference value; (d) is a stated research direction. We recommend the reader hold the paper to (a) and (b) as the contribution and to (c) and (d) as scaffolding.

8. Discussion, Open Questions, Future Work

8.1. **Immediate applications.** The applications companion paper (in preparation) will develop five applications enabled by this foundation: *(a)* adversarial reflexive ISG on prover–verifier interactive protocols (the "epistemic body as counterexample-oracle" scenario of Section 5.6), together with the defensive stack (transcript / statement / semantic / bridge-resistance binding); *(b)* the *semantic-body ↔ epistemic-body triangulation methodology* (Remark 71) for the disciplined introduction of new concepts into a multi-agent programme's shared reference space — given a target concept c, the semantic body's tuning-parametric substrate and the epistemic body's multi-agent shared substrate jointly discipline the introduction, with substrate coherence between the two as the programme-referenceability gate; *(c)* technological transfer as substrate DIA between source and target domains; *(d)* the two-routes dissociation of semantic state-change (autonomy vs contact-absorption) as a substrate structure result on the two-register dynamics of Section 5.7; *(e)* monitor-of-monitor scenarios in which a monitor's own admissibility is the operand of a further ISG pass.

8.2. **Open structural questions.** The following remain open at the framework level.

(Q1) The full class over which Remark 12 holds, together with its algebraic form. Candidate approaches include measure-preserving atomic decomposition (Solid → Atomic) and reflective subcategory embeddings (Atomic → Solid).

(Q2) Conjecture 62 beyond the Σ^ω witnessed instance. The obstacle is exhibiting, for a general Atomic-Geometry spectrum, an intrinsic clopen characterisation of DIA-freedom.

(Q3) The universal-substrate question: does the category of grammatical interpretations of a fixed statement space $\mathcal{S}$ admit an initial or terminal object (canonical minimal, canonical maximal spectrum)?

(Q4) The multi-substrate coherence problem: under what conditions do candidate grammars $G_1, G_2, \ldots$ converge on the same Σ up to isomorphism? This is the framework-level counterpart of the classical theorem-selection problem.

(Q5) Observable completeness: does every pointing DIA correspond to a witnessable substrate DIA under some grammar-extension? An affirmative answer would tighten the duality between the two indeterminacy modes.

(Q6) Two routes for semantic state-change: the two-register dynamics of Section 5.7 suggests a bipartition of the mechanisms by which a state can move on Σ_{state} — an *internal-autonomy* route (driven purely by the monotone accumulator crossing an internal threshold) versus a *contact-absorption* route (driven by an external register imposing a grammar-extension). Whether this bipartition is exhaustive and how the two routes interact on mixed dynamics is open.

(Q7) Independence of the separating criteria of Table 1: are the listed criteria pairwise irreducible, or does some pair collapse under a suitable reduction? The candidate collapses to rule out first are Incompleteness against Inaccuracy, and Incoherence against Inconsistency, which differ in the catalogue by the side of the presentation they test rather than by an established separation result.

(Q8) Sufficiency of the adapters of Table 2: is every criterion one might add to Table 1 applicable under each DIA operator after supplying one of A1–A5, or do further adapter classes exist?

(Q9) Completeness of Table 3: can it be shown that every well-typed quadruple of criterion, operator, direction and route is an instance of a recorded pairing? A positive answer would upgrade the table from a witnessed catalogue to a classification theorem, which is the principal formal debt of Section 4.7.

8.3. **Grounding for downstream programme work.** ISG was extracted from patterns already present in independently developed programme substrates. Its role forward is dual: it grounds the semantic-body / epistemic-body work (Section 5.5, Section 5.6) as an unified two-pole substrate discipline, and it provides the applications companion with a spectrum-grammar vocabulary in which several previously-informal moves acquire a formal home. The coordinated-release strategy for the ISG cluster (this paper, the applications companion, the MCK foundations, the refinement / finite-confirmation companion, the semantic-body manuscript) is designed to make these grounding relations citable simultaneously.

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